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A many-objective evolutionary algorithm based on interaction force and hybrid optimization mechanism

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ABSTRACT

In many-objective optimization, both convergence and diversity are equally important. However, in high-dimensional spaces, traditional decomposition-based many-objective evolutionary algorithms struggle to ensure population diversity. Conversely, traditional Pareto dominance-based many-objective evolutionary algorithms face challenges in ensuring population convergence. In this paper, we propose a novel many-objective evolutionary algorithm based on interaction force and hybrid optimization mechanism (MaOEAIH) for effectively addressing the difficulty in balancing convergence and diversity. First, we use the concept of interaction force to simulate the convergence (akin to gravity) and diversity (repulsion) of the population. Subsequently, we design an optimization mechanism that combines decomposition and Pareto dominance to enhance the convergence and diversity of the population separately. Simultaneously, to eliminate dominance resistance solutions, we propose a quartile method based on boundary solutions. Additionally, Random perturbations are also introduced to certain individuals within the population to facilitate their escape from local optima. MaOEAIH is compared with some state-of-the-art algorithms on 31 well-known test problems with 3-15 objectives. The experimental results show that, compared to other algorithms, MaOEAIH not only obtains solution sets of higher quality when dealing with different types of many-objective optimization problems, but also effectively addresses key challenges including insufficient selection pressure, difficulty balancing convergence and diversity, and susceptibility to population entrapment in local optima within many-objective optimization scenarios.

1. Introduction

In reality, there are many optimization problems, such as sensor configuration [1], design of robot neuro-motion-controllers [2], and job-shop scheduling [3], in which in most cases there are two or more conflicting objectives. Improving the performance of one objective is likely to lead to a decrease in the performance of other objectives. Therefore, it is necessary to optimize these conflicting objectives as much as possible to achieve the optimal state of the overall objective. When the number of objectives in these problems is less than three, they are called multi-objective problems (MOPs) [4,5]. Otherwise, they are called many-objective problems (MaOPs) [6]. A MOP (MaOPs) can be expressed by (1):

$$\begin{cases} Minimize\ F(x) = \left(f_1(x), f_2(x), \ldots, f_M(x)\right)^T \\ Subject\ to\ x \in \Omega \end{cases} \tag{1}$$

where $x = \left(x_1, x_2, \ldots, x_n\right)^T$ is an n-dimensional decision vector in the decision space Ω. M is the number of objectives, and $F(x)$ is an M-dimensional objective vector. For two given decision vectors x_a and x_b, if x_a dominates x_b ($x_a \prec x_b$), then $\forall i \in \{1, 2, \ldots, M\}$, $f(x_a)_i \leq f(x_b)_i$ and $\exists i \in \{1, 2, \ldots, M\}$, $f(x_a)_i < f(x_b)_i$. A solution that is not dominated by any other solutions is called a non-dominated solution. The set of non-dominated solutions is called the Pareto-optimal set (PS), and the graph formed by the Pareto optimal solutions is called the Pareto-optimal front (PF).

As the number of objectives in MaOPs increases, the performance of traditional multi-objective evolutionary algorithms (MOEAs) in solving MaOPs gradually deteriorates, making it difficult to maintain a balance between population diversity and convergence [7], and solving MaOPs becomes increasingly challenging. To effectively address MaOPs, many excellent many-objective evolutionary algorithms (MaOEAs) [8–10] have been proposed. These algorithms can be roughly categorized into four types:

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2210-6502/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

The first category is Pareto dominance-based MaOEAs. These algorithms generally employ two strategies for individual selection. The first strategy is to select individuals by modifying the Pareto dominance relation, such as GPO dominance [11], CSDR-dominance [12], and localized dominance [13]. These methods essentially relax the dominance relationship, allowing for the comparison of solutions that cannot be directly compared, thus maintaining selection pressure. Although these methods enhance the selection pressure of MaOEAs, they heavily rely on parameter settings and often struggle to find suitable parameters. The second strategy is to select individuals through novel diversity selection criteria. For instance, Xu et al. improve the selection pressure by choosing solutions with the best convergence index value [14]. Zhou et al. utilize a diversity maintenance mechanism based on global and local diversity to manage the population's diversity [15]. Chen et al. enhance the diversity of the population on different types of PFs through a diversity ranking method [16]. While these methods are effective in increasing selection pressure, they may fall slightly short in balancing diversity and convergence.

Decomposition-based MaOEAs belong to the second category, where the principle of these algorithms is to decompose an MaOPs into several subproblems using a set of weight vectors and simultaneously optimize these subproblems. MOEA/D [17] is a representative algorithm of this category. However, for decomposition-based MaOEAs that use fixed weights, the initial problem is that the number of individuals in the population must be the same as the number of weight vectors. Moreover, when faced with irregular-shaped PF such as disconnected, degenerate [18], reversed [19,20], constrained, severely scaled [21], and complex fronts [22], the performance of the algorithm deteriorates. This deterioration leads to the inability to generate uniformly distributed solutions and also results in wasted weight vectors. To address this problem, several methods for adaptive weight vectors have been proposed. For example, SRV [23] applies an improved k-means clustering method to extract weight vectors from the population. DEAdas [24] utilizes a dual-adjustment strategy to adjust the weight vectors. In MOEA/D-UR [25], a metric that detects improvements is employed to adjust the weight vectors when necessary. However, even though adjusting the weight vectors enables the decomposition-based MaOEAs to handle problems with irregular PF effectively, the number of weight vectors is often not arbitrary, and designing a reasonable and effective weight generation strategy still poses certain difficulties.

Indicator-based MaOEAs form the third category. Performance evaluation indicators can evaluate the comprehensive performance of candidate solutions, so introducing performance indicators can select candidate solutions and transform MaOPs into single-objective problems judged by performance indicators. Algorithms such as HyPE [26], MaOEA/IGD [27] and R2HCA-EMOA [28] use accurate indicators or estimated indicator values such as hypervolume(HV) [29], inverted generational distance(IGD) [30], and R2 [31] to guide the selection process of many-objective optimization. However, the computation cost of accurate indicator values is very expensive for indicator-based many-objective optimization algorithms. Secondly, the solutions generated are difficult to ensure the uniformity of distribution. Finally, it is still a difficult problem to select the most suitable reference point for various algorithms of this type.

The fourth category of MaOEAs is based on alternative evolutionary mechanisms. These algorithms employ unique evolutionary mechanisms distinct from the three common mechanisms mentioned above. For instance, LM-DAS [32], based on the duality analysis of the objective space and decision space, reduces problem complexity by prioritizing resource allocation to variables that are relevant to important objectives. However, incorrect resource allocation may lead to instability in the results. Another example is S3-CMA-ES [33], which utilizes a series of small subpopulations to approximate the PF, thereby improving the evolutionary speed of the population. However, the population is prone to getting trapped in local optima and fails to converge to the PF. Additionally, DVC-MOEA [34], based on decision variable classification, adopts a novel method to generate promising solutions. However, variations in the contribution of decision variables can lead to incorrect classification, thereby affecting the algorithm's performance. Du et al. [35] have bridged evolutionary algorithms and network science, proposing a networked evolutionary algorithm framework that significantly enhances the performance of evolutionary algorithms. Nonetheless, the complexity of the network structure may lead to increased implementation difficulty. Fister et al. [36] integrated novelty search with the differential evolution algorithm, enhancing the algorithm's global optimization performance. However, this approach requires careful parameter tuning.

In summary, existing MaOEAs exhibit several advantages, such as effectively maintaining diversity or convergence in high-dimensional spaces, generating a set of optimal solutions in a single run, deriving global approximate optimal solutions without requiring additional assumptions, high universality, strong robustness, and fast computational speed, among others. These attributes enable them to perform well in handling MaOPs. However, there are still some shortcomings. Firstly, in decomposition-based MaOEAs, the methods for adjusting weight vectors cannot effectively handle MaOPs with both regular and irregular PF simultaneously. Secondly, most MaOEAs, to some extent, prioritize either convergence or diversity, and fail to strike a good balance between the two. Thirdly, In Pareto-dominance-based MaOEAs, the presence of dominance resistance solutions [37] reduces the selection pressure of the algorithm, severely affecting the evolution of the population. Fourthly, In high-dimensional spaces, the search capability of evolutionary operators in MaOEAs is greatly constrained, making the population get trapped in local optima easily.

Inspired by the fact that physical systems with multiple bodies and interactions in nature tend to reach an ordered and homogeneous state, and the applicability of MaOEAs based on decomposition and Pareto dominance in various problems [38], we propose a many-objective evolutionary algorithm based on interaction force and hybrid optimization mechanism aiming to address the four aforementioned problems. Specifically, we use interaction forces in the Pareto dominance optimization mechanism to select a population that is close to the PF and roughly conforms to it, and then use this population as weight vectors in the decomposition-based optimization mechanism to emphasize the convergence of the population. Next, we use interaction forces again to select a population with good diversity. While using the quartile method based on boundary solutions to eliminate dominance resistance solutions, ensuring the convergence of the solutions. Finally, we use the obtained population as weight vectors to guide the population search for solutions that exhibit both convergence and diversity. Additionally, when the population gets trapped in local optima, we apply random perturbation to some individuals. The main contributions of this paper are as follows:

- This paper proposes Gravity and Repulsion-Interaction Force as measures for assessing the diversity of populations and selecting populations with good convergence and diversity as weight vectors.
- A hybrid optimization mechanism that combines decomposition and Pareto dominance is proposed to enhance the convergence and diversity of populations, while maintaining a balance between the two.
- We employ the quartile method and the boundary solutions of the population to roughly estimate the range of PF to identify solutions outside this range as dominance resistance solutions and eliminate them.
- Some offspring will be generated by randomly perturbing certain alleles of the parents, aiming to increase the probability of exploring regions far from the parent's area, thus allowing the population to escape local optima.

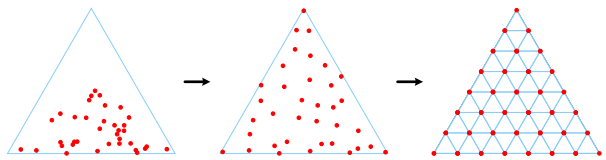

Fig. 1. Particles gradually transition from a chaotic and disordered state to a uniform and ordered state under the influence of Riesz's s-energy.

The remainder of this paper is organized as follows. In Section 2, we will review related work. In Section 3, we will provide a detailed description of our new algorithm. Then, experimental settings and results will be presented in Section 4. Finally, in Section 5, we summarize this paper and discuss future work.

2. Related work

In this section, we will introduce the concept of interaction force and the principle of Riesz s-energy, followed by an explanation of the quartile method.

2.1. Interaction forces and Riesz s-energy

Interaction forces [39] refer to the forces that an object exerts on other objects while also being subject to forces exerted by those objects. The effects of such forces may lead to a state of static or dynamic equilibrium between objects. Common types of Interaction forces include gravities, repulsions, elastic forces, and nuclear forces, among others.

Riesz s-energy [40] is essentially a type of energy that describes the interaction between particles and also represents repulsion, a type of interaction force. The principle behind it is that in natural physical systems consisting of multiple particles and interactions, the system will eventually reach a state of minimum total potential energy. In brief, by continuously selecting the point farthest from other points in the point set, a set of uniformly spaced points will eventually be obtained. In the homogeneous particle equilibrium system shown in Fig. 1, the initial distances between the particles are very close to each other, and because of the strong repulsion between them, they begin to separate from each other, making the distance between them increase gradually while their potential energy decrease at the same time. Eventually, the particles will reach an equilibrium state, and distribute uniformly on the plane. The mathematical definition of Riesz s-energy is as follows:

$$E = \sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} \frac{1}{\left\| x_i - x_j \right\|^s} \quad x \in R \tag{2}$$

where E represents the total energy of all points, and energy is represented by the Euclidean distance between two points, such that the greater the distance, the smaller the energy and the more uniform the points. n is the number of points, x_i and x_j represent the ith and jth points of the point set R, and $s \geq 0$ is a parameter that controls the emphasis on uniform distribution.

Recently, some articles in the field of many-objective optimization have also used Riesz s-energy. For example, in [41], the author used Riesz s-energy to select a set of points of a specified size and uniformity from the test suits to test the performance of MaOEAs. In [42], the author used Riesz s-energy to generate initially uniform reference points in the unit simplex. However, the application of Riesz s-energy in the field of many-objective optimization is still relatively limited, and it has great potential for development. With reasonable improvements, Riesz s-energy can be applied to select individuals with good convergence and diversity in many-objective optimization.

2.2. Quartile method

The quartile method [37] is a statistical technique used to describe the distribution of a dataset. It is commonly employed in exploratory data analysis to effectively identify the central tendency and dispersion of the dataset. Compared to traditional statistical measures such as mean and standard deviation, the quartile method exhibits stronger robustness and can effectively handle the influence of outliers and extreme values. By computing the interquartile range (IQR) of the dataset and using multiples of the IQR, the quartile method enables the determination of threshold values for detecting outliers. The quartile method is defined as:

$$\begin{cases} Q_1 = \mathbf{X}_{\left\lfloor \left(\frac{\|\mathbf{X}\|}{4} \right) \right\rfloor + 1} \left(\frac{\|\mathbf{X}\|}{4} - \left\lfloor \left(\frac{\|\mathbf{X}\|}{4} \right) \right\rfloor \right) + \\ \quad \mathbf{X}_{\left\lfloor \left(\frac{\|\mathbf{X}\|}{4} \right) \right\rfloor} \left(1 - \left(\frac{\|\mathbf{X}\|}{4} - \left\lfloor \left(\frac{\|\mathbf{X}\|}{4} \right) \right\rfloor \right) \right) \\ Q_3 = \mathbf{X}_{\left\lfloor \left(\frac{3\|\mathbf{X}\|}{4} \right) \right\rfloor + 1} \left(\frac{3\|\mathbf{X}\|}{4} - \left\lfloor \left(\frac{3\|\mathbf{X}\|}{4} \right) \right\rfloor \right) + \\ \quad \mathbf{X}_{\left\lfloor \left(\frac{3\|\mathbf{X}\|}{4} \right) \right\rfloor} \left(1 - \left(\frac{3\|\mathbf{X}\|}{4} - \left\lfloor \left(\frac{3\|\mathbf{X}\|}{4} \right) \right\rfloor \right) \right) \end{cases} \tag{3}$$

$$\begin{cases} UB = Q_3 + k\left(Q_3 - Q_1\right) \\ LB = Q_1 - k\left(Q_3 - Q_1\right) \end{cases} \tag{4}$$

where Q_1 is the first quartile, Q_3 is the third quartile, X is a set of ordered one-dimensional data, LB is the lower bound, UB is the upper bound, $k = 1.5$ indicates moderately extreme outliers, and $k = 3$ indicates highly extreme outliers.

3. Proposed algorithm

First, we will introduce the overall framework of MaOEAIH. Then, we will elaborate on our proposed Gravity and Repulsion-Interaction Force method and the process of using it to select individuals and weight vectors. Next, we will provide a detailed description of the hybrid evolution mechanism. Afterwards, we will explain the process of eliminating domination resistance. Finally, we will describe our mutation operation in detail.

3.1. The overall framework of MaOEAIH

Algorithm 1 presents the overall framework of our proposed MaOEAIH. First, we initialize a population **P** of size N and M objectives. Simultaneously, we initialize the parameters C, V, H, and **HP**, and the details of these parameters, as well as those mentioned subsequently, are described in Table 1. Then, the population **P** starts to iteratively evolve. In each generation, a random subset of parents $\mathbf{P}'$ is selected from **P** to generate offspring **F** through crossover and mutation or perturbation. It is worth noting that our algorithm is different from most MaOEAs in that we also apply a perturbation operation to **F** to help the population escape from local optima. Finally, we employed the interaction forces and hybrid optimization mechanism to select N individuals from a total of $2N$ individuals in $\mathbf{P}'$ and **F** as the next generation population. Simultaneously return the next generation of **PSET**. This evolution process is repeated until the stopping criterion is met. In the following sections, we will describe the main components of MaOEAIH in detail.

Algorithm 1 Framework of the Proposed MaOEAIH

Input: N (Population size), M (Number of objectives)
Output: **P** (Population)
1: **P** ← Initialize Population(N, M)
2: $V, C \leftarrow 0$
3: $H \leftarrow$ inf
4: **HP** ← **P**
5: **PSET** ← V, C, H, **HP**
6: **while** termination criterion not fulfilled **do**
7: **P**$'$ ← MatingSelection(**P**)
8: **F** ← VariationOrPerturbation(N, **P**$'$, V)
9: **P**, **PSET** ← MixedSelection(N, M, **P**, **F**, **PSET**)
10: **end while**

Table 1
Pseudocode common variables.

Variable	Description	Type
C	Parameter that determines whether the current population should employ an optimization mechanism based on decomposition or one based on Pareto dominance	Scalar
V	Parameter that decides if the current population needs perturbation	Scalar
H	Records the optimal values of population convergence and diversity	Scalar
HP	Stores historical populations	Set
PSET	A set of variables containing C, V, H, **HP**	Set
P	Parent population	Set
F	Offspring population	Set
P$'$	A mixed population of offspring and parent populations	Set

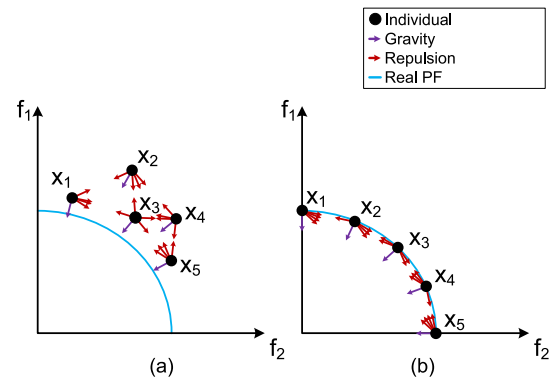

Fig. 2. Under GaRIF's influence, individuals repel each other and are gradually attracted towards PF, transitioning from (a) initial chaotic state to (b) final state of high diversity and convergence.

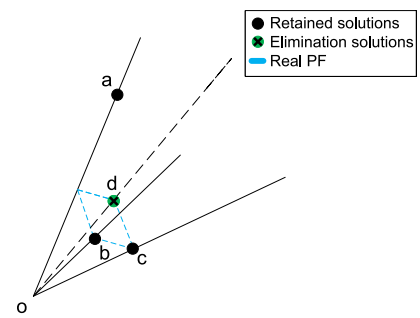

Fig. 3. Using the Manhattan distance to calculate the repulsion in the case of the same angle between individuals will tend to select individuals far from the PF.

3.2. Gravity and repulsion-interaction force

The PF in MaOPs is a high-dimensional hypersurface, making the task of maintaining solution diversity a challenging endeavor. Inspired by the effectiveness of Riesz s-energy in promoting point set uniformity, we incorporate it into a Pareto dominance-based MaOEAs as a criterion for diversity preservation. However, as the number of objectives in MaOPs increases, the Pareto dominance relationship gradually loses its effectiveness, leading to a deterioration in population convergence. Furthermore, Riesz s-energy, being a repulsion, tends to favor individuals with larger Euclidean distances from other individuals, which can result in the selection of individuals far away from the PF and further worsen population convergence.

To address these issues, we introduce the concept of gravity based on the inspiration from interaction forces and enhance the Riesz s-energy (repulsion) by employing different distance metrics. Finally, we propose the Gravity and Repulsion-Interaction Force (GaRIF), aiming to improve population convergence while maintaining diversity. Specifically, GaRIF is a method that simulates gravity and repulsion to drive population individuals closer to the PF while ensuring uniform diffusion across the entire PF. To further clarify GaRIF, a conceptual illustration is provided in Fig. 2.

Fig. 2(a) illustrates that initially, population individuals are in a chaotic and disordered state, with repulsion acting between each pair of individuals, pushing them away from each other. At the same time, they are also subject to gravity from the PF, pulling them closer to the PF. Over time, the distance between individuals increases, the repulsion gradually decreases, reaching its minimum value, no longer forcing individuals away from each other but maintaining a state of equilibrium. Similarly, as the gravity attracts individuals to the PF, the distance between individuals and the PF also reaches its minimum value, causing individuals to stop moving and remain in a state of equilibrium. As shown in Fig. 2(b), ultimately, these two forces will cause individuals to approach the PF infinitely while uniform spreading across the entire PF, achieving an orderly balanced state for the population. In other words, GaRIF maintains population diversity by simulating repulsion and ensures population convergence by simulating gravity. Its specific expression is shown in Eqs. (5):

$$\begin{cases} E_{GaRIF} = G \cdot R \\ G = \left(\sum_{i=1}^{n} \sum_{k=1}^{m} \left| x_k^i \right| \right)^{h_1} \\ R = \sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} (MD \cdot CD)^s \\ MD = \left(\dfrac{1}{\sum_{k=1}^{m} \left| x_k^i - x_k^j \right|} \right)^{h_2} \\ CD = \left(\dfrac{1}{1 - \cos\left(x^i, x^j\right)} \right)^{h_3} \\ x \in P \end{cases} \tag{5}$$

where n is the population size, m is the number of objectives, **P** is the current population, x_k^i is the kth objective value of the ith individual in **P**, the parameter s ($s \geq 0$) controls the uniform distribution and is set as $m-1$, following the Ref. [41], h_1 is the weight coefficient of the gravity term, h_2 is the weight coefficient of the reciprocal of the Manhattan distance between individuals, and h_3 is the weight coefficient of the reciprocal of the difference between one and Cosine similarity between individuals. G represents the Manhattan distance from all individuals in the population to the origin. Since in minimization problems a smaller G value indicates better convergence of the population [43], it can be used as gravity to maintain the convergence of the population. R is the

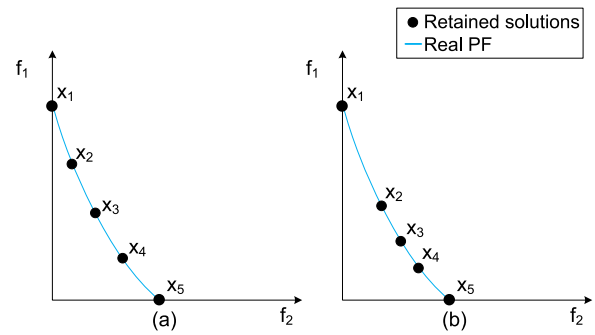

Fig. 4. Using (b) Cosine similarity to calculate repulsion in problems with convex PF results in uneven distribution of individuals, while adopting (a) Manhattan distance for repulsion calculation maintains a uniform distribution of individuals.

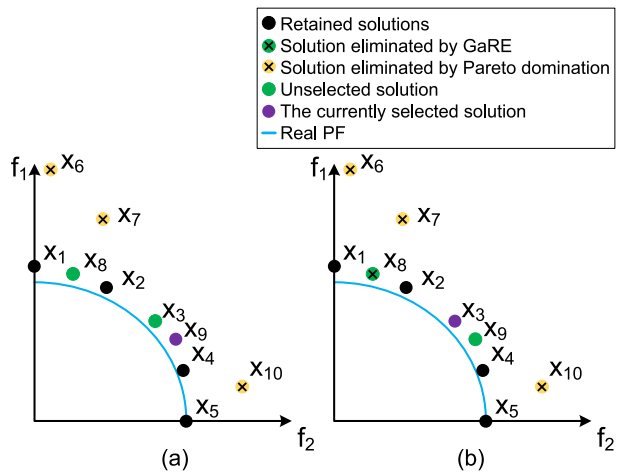

Fig. 5. (a) Roughly select a population with the minimum GaRIF by choosing individuals with the current minimum GaRIF. (b) Achieve a more precise selection of a population with the minimum GaRIF by sequentially replacing individuals obtained in (a) with the remaining individuals.

sum of the reciprocal s powers of the distance between each individual in the population and other individuals, representing repulsion used to maintain population diversity. The smaller R, the greater the distance between individuals in the population, and the better the diversity of the population. E_{GaRIF} represents GaRIF, which is a product of population gravity and repulsion. Therefore, a smaller E_{GaRIF} value indicates better balance between diversity and convergence within the population.

It is worth noting that the calculation of the repulsion term R, differs from the computation of Riesz s-energy. Instead of using $1/\|x_i - x_j\|$ in Riesz s-energy, we employ $MD \cdot CD$. This substitution is motivated by the following reasons: Firstly, in high-dimensional spaces, using Euclidean distance to measure the distance between points is not sufficiently accurate [44]. Therefore, we substitute Euclidean distance with Manhattan distance and Cosine similarity. Secondly, as shown in Fig. 3, in the case where the angles between individuals are the same, calculating R using Manhattan distance would select individual a with a large Manhattan distance to other individuals, instead of individual d, which is closer to the PF. Employing Cosine similarity might preserve d, but it may fail to ensure diversity, as demonstrated in Fig. 4(b), where using Cosine similarity to compute R could select individuals with equal angles within the convex PF, but the distances between them are not equal. Conversely, in Fig. 4(a), calculating R with Manhattan distance results in equal distances between the selected individuals. To combine the advantages of Manhattan distance and Cosine similarity, we use h_2 and h_3 to represent their respective proportions in computing

Algorithm 2 GaRIFSelection1

Input: N (Population size), M (Number of objectives), **P** (Population), **F** (Offspring), h_1 (Control the parameters of gravity), h_2, h_3 (Control the parameters of repulsion)
Output: **P** (Population), E_{GaRIF} (GaRIF of **P**)

To perform non-dominated sorting on $\mathbf{P} \cup \mathbf{F}$, obtaining **Q** (the last non-dominated layer individuals) and $\mathbf{P}'$ (non-dominated layers individuals except the last layer)
$\mathbf{GM} \leftarrow$ GaRIFMatrix$(N, M, \mathbf{P} \cup \mathbf{F}, h_2, h_3)$
for $i = 1 \rightarrow (N - \mathbf{P}')$ **do**
  for $j = 1 \rightarrow |\mathbf{Q}|$ **do**
    Calculate E_{GaRIF} of the current population using Eqs (5), h_1, and **GM**, and store this value in $\mathbf{EM}(j)$
  end for
  Add the individual a corresponding to the minimum value in $\mathbf{EM}(j)$ to $\mathbf{P}'$, and remove a from **Q**
  $\mathbf{O} \leftarrow \mathbf{a} \cup \mathbf{O}$
end for
for $i = 1 \rightarrow |\mathbf{O}|$ **do**
  Calculate E_{GaRIF} of $\mathbf{P}'$ using Eqs (5), h_1, and **GM**
  for $j = 1 \rightarrow |\mathbf{Q}|$ **do**
    Calculate E_{GaRIF} of the current population using Eqs (5), h_1, and **GM**, and store this value in $\mathbf{EM}(j)$
  end for
  $[\mathbf{EM}_{min}, l] \leftarrow \mathbf{EM}$
  if $E_{GaRIF} > E^{min}_{GaRIF}$ **then**
    $\mathbf{P}' \leftarrow \mathbf{P}' \cup \mathbf{Q}(l) - \mathbf{O}(i)$
    $\mathbf{Q} \leftarrow \mathbf{Q} - \mathbf{Q}(l) + \mathbf{O}(i)$
    $E_{GaRIF} \leftarrow E^{min}_{GaRIF}$
  end if
end for
$\mathbf{P} \leftarrow \mathbf{P}'$

Algorithm 3 GaRIFMatrix

Input: N (Population size), M (Number of objectives), **P** (Population), h_2, h_3 (Control the parameters of repulsion)
Output: **GM** (The distance matrix between two individuals in the storage population)

Calculate the difference between the one and Cosine similarities of each pair of individuals in **P**, and store the results in the matrix **OneCos**.
for $i = 1 \rightarrow N$ **do**
  for $j = 1 \rightarrow N$ **do**
    if $i = j$ **then**
      continue
    end if
    for $l = 1 \rightarrow M$ **do**
      $u_1 \leftarrow u_1 + |f(i)_l - f(j)_l|$
    end for
    $\mathbf{GM}(i,j) \leftarrow \frac{1}{((u_1)^{h_2} \cdot (\mathbf{OneCos}(i,j))^{h_3})^{M-1}}$
  end for
end for

R. Finally, since the distance measure should be inversely proportional to R, we calculate the Cosine similarity by subtracting the Cosine distance from 1.

The search space for finding a population of size N with the minimum GaRIF value from a set containing $2N$ individuals comprising both parents and offspring, is $\binom{2N}{N}$ [41], which incurs a significantly

high time complexity. To address this issue, this paper employs a simplified approach for selecting populations of size N with the smallest possible GaRIF value.

As shown in Algorithm 2, if the selected non-dominated individuals do not fulfill the population size after non-dominated sorting, the individuals with the minimum GaRIF value among the remaining non-dominated individuals and the already selected individuals are chosen to form the population. This process resembles a greedy algorithm (steps 3–9 of Algorithm 2). For example, in Fig. 5(a), if 5 individuals need to be chosen and assumes that x_9 has a smaller GaRIF value with the selected individuals compared to any other unselected individual, then it will be selected. However, this selection may not be accurate enough. Therefore, we individually replace the selected individuals chosen by GaRIF with the unselected non-dominated individuals. If the replacement does not decrease the GaRIF value of the population, the replacement is canceled (Steps 10–20 of Algorithm 2). For example, in Fig. 5(b), if replacing x_9 with x_3 decreases the GaRIF value of the population, x_3 will be retained. Although the method cannot find populations with the smallest GaRIF values but populations with the smallest possible GaRIF values, its time complexity is only $O(N^2)$. Additionally, if the GaRIF value is computed for each individual during the selection process, the total time complexity of this method would be $O(N^4M)$. To avoid redundant calculations, a matrix is employed to store the distances between previously computed individuals, similar to the approach described in [41]. This reduces the overall time complexity to $O(N^3M)$, as shown in Algorithm 3.

3.3. Hybrid optimization mechanism

Although GaRIF can improve the convergence and diversity of the population, there is still room for improvement for problems that are difficult to converge. Decomposition-based MaOEAs can effectively converge to the PF. Therefore, the population selected by GaRIF can be used as the weight vector for decomposition-based MaOEAs (this paper adopts MOEA/D) to guide the search for solutions and further improve convergence. Fig. 6(a), (b), and (c) illustrate this process. However, balancing gravity and repulsion is not straightforward, which leads to a lack of diversity in the population selected by GaRIF. For example, in Fig. 6(c), the distance between x_3 and x_4 is significantly smaller than the distances between other individuals. It is worth noting that the population selected through the above steps has already approximated the range of the PF. By constraining the offspring within this range (Section 3.4), we can eliminate gravity and its impact on diversity, select a population with good diversity through GaRIF, and avoid the concern that repulsion and dominance resistance solutions deviate the population from the PF which significantly reduces convergence, as shown in Fig. 6(d). However, the determined range of the PF is only an approximate range, and the population may still deviate slightly from the PF, leading to reduced convergence of the population. To address this, we once again use the population as the weight vector to guide the search for solutions and obtain a population with both good convergence and diversity, as shown in Fig. 6(e).

Furthermore, Algorithm 4 presents the implementation and details of the above steps and provides the transition conditions of the optimization mechanism and the setting of some coefficients. Since the initial value of C is 0, Algorithm 4 first adopts a Pareto dominance-based optimization mechanism to evolve the population. (steps 1 to 2 of Algorithm 4). When the historical minimum value of GaRIF in the population remains unchanged for a certain number of generations and the population has already undergone perturbation, C is set to 1 or 3 (steps 12 to 31 of Algorithm 4). Subsequently, we employ a decomposition-based optimization mechanism as a replacement for the Pareto dominance-based mechanism(steps 3 to 8 of Algorithm 4). This substitution is motivated by the stability of the GaRIF value in the population, which signifies a certain degree of stability in both convergence and diversity. Similarly, when the historical minimum value of the sum of distances from each individual to the origin remains constant for a specified number of generations and the population has already undergone perturbation, C is set to 2 (steps 12 to 31 of Algorithm 4), and we switch back to the Pareto dominance-based optimization mechanism (steps 9 to 10 of Algorithm 4). This constancy in the minimum sum of distances from the origin represents a certain level of stability in population convergence. The "a certain number of generations" will be introduced in Section 4.6. Moreover, in the initial usage of the Pareto dominance-based optimization mechanism, we consider both gravity and repulsion to be equally important. Therefore, we set the GaRIF coefficients, h_1, h_2, and h_3, to 1, 1, 1. However, in the second usage of the Pareto dominance-based optimization mechanism, we prioritize the diversity of the population. Thus, we set the GaRIF coefficients, h_1, h_2, and h_3, to 0, 1, 0.

Algorithm 4 MixedSelection

Input: N (Population size), M (Number of objectives), **P** (Population), **F** (Offspring), **PSET** (The parameter set containing the parameters V, C, H, and **HP**)
Output: **P** (Population), **PSET** (The parameter set containing the parameters V, C, H, and **HP**)

```
1: if C = 0 then
2:     P, E_GaRIF ←GaRIFSelection1(N, M, P, F, 1, 1, 1)
3: else if C = 1 || C = 3 then
4:     Using the population obtained from a Pareto dominance-based optimization mechanism as the weight vector W.
5:     Set the ideal point as the coordinate origin
6:     The next generation population P' was selected from P and F using the same method as MOEA/D
7:     P← P'
8:     Calculate the Manhattan distance from the origin to each individual in P, and store the results in E_GaRIF
9: else if C = 2 then
10:    P,E_GaRIF ←GaRIFSelection2(N, M, P, F, HP, 0, 1, 0)
11: end if
12: if E_GaRIF < H || (C = 1 and E_GaRIF < H + 10^-6) then
13:    H ← E_GaRIF
14: else
15:    if H remains unchanged for a certain number of generations then
16:        if V = 0 then
17:            V ← 1 /*Start perturbation*/
18:            Counting the number of generations with unchanged H from 0
19:        else
20:            if C = 0 then
21:                C ← 1
22:            else if C = 1 then
23:                C ← 2
24:                HP← P
25:            else if C = 2 then
26:                C ← 3
27:            end if
28:            Counting the number of generations with unchanged H from 0 and V is set to 0
29:        end if
30:    end if
31: end if
32: PSET← V, C, H, HP
```

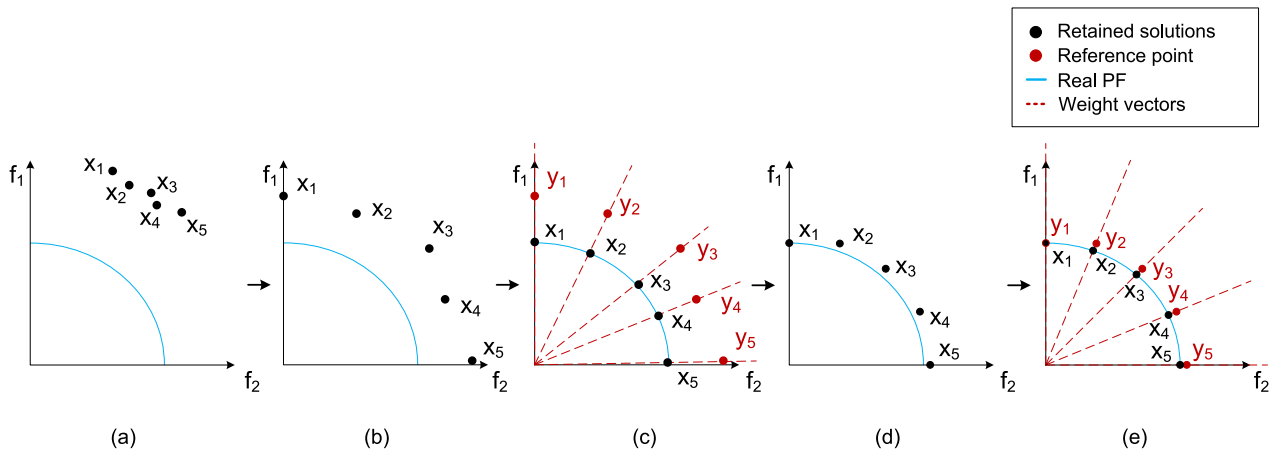

Fig. 6. Hybrid optimization mechanism. The population transitions from (a) to (b) and from (c) to (d) using a Pareto dominance-based optimization mechanism to enhance diversity. The shift from (b) to (c) and (d) to (e) employs a decomposition-based optimization mechanism to enhance convergence.

3.4. Eliminating dominance resistance solution

Algorithm 5 SearchDRS

Input: N (Population size), M (Number of objectives), $\mathbf{P}(x)$ (Population individual x), **HP** (Historical population)
Output: *DRS* (Dominance Resistance Solution Discriminant Parameters)

1: Select the M individuals closest to the M coordinate axes from **HP** as the boundary solutions **Bs**
2: **for** $i = 1 \rightarrow N$ **do**
3: $\quad$ **HPS**$(i) \leftarrow$sum(**HP**(i))
4: **end for**
5: Sort **HPS** in ascending order.
6: **if** $N \geq 4$ **then**
7: $\quad$ Calculate Q_1, Q_3 and UB by Eqs. (3) and (4)
8: **end if**
9: **for** $i = 1 \rightarrow M$ **do**
10: $\quad$ **BsS**$(i) \leftarrow$sum(**Bs**(i))
11: **end for**
12: $maxUB \leftarrow$ max(**BsS**(i))
13: **if** $maxUB > UB$ **then**
14: $\quad$ $UB \leftarrow maxUB$
15: **end if**
16: **if** sum(**P**(x)) **then** $> UB$
17: $\quad$ $DRS \leftarrow$ true
18: **else**
19: $\quad$ $DRS \leftarrow$ false
20: **end if**

In order to make the generated offspring as close as possible to the PF, we employ the quartile method based on boundary solutions to eliminate dominance resistance solutions and restrict the range of offspring generation. In Algorithm 5, we first save the final population **HP** obtained from the initial use of the decomposition-based optimization mechanism. Next, we calculate the sum of each individual on each objective value and store it in *HPS*. Subsequently, we compute the upper bound UB of *HPS* using formulas (3) and (4), and then use UB as the upper bound for the PF. In steps 16–20 of Algorithm 5, by comparing the sum of each objective value in the offspring individual with UB, We restrict offspring generation to the vicinity of PF. If the sum of each objective value in the offspring exceeds UB, it is considered a dominance resistance solution and needs to be eliminated. Otherwise, it is retained.

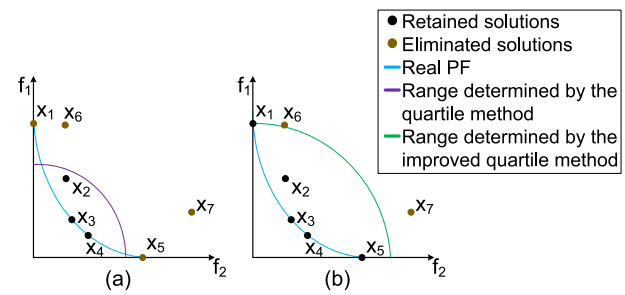

Fig. 7. (b)The improved quartile method expands the detection range near boundary solutions compared to (a) the unimproved quartile method.

In this paper, according to [37], the parameter k in the quartile method is set to 1.5. However, if the distribution of the **HP** population is uneven and the value of k is small, as shown in Fig. 7(a), the dense region in the middle and sparse regions on the sides, there is a possibility of erroneously identifying boundary solutions outside the purple region as dominance resistance solutions and eliminating them, although these solutions are not actually dominance resistance. This not only fails to improve the algorithm's performance but may also reduce it. To address this issue, in the first step of Algorithm 5, we select the M individuals closest to the M coordinate axes from **HP** as the boundary solutions. Then, in steps 9–12, we calculate the maximum sum of objective values, $maxUB$, among these M individuals and compare it with UB. If $maxUB$ is greater than UB, we replace UB with $maxUB$ to expand the range of offspring generation and avoid erroneously eliminating useful solutions, as shown in Fig. 7(b).

Algorithm 6 provides specific steps for eliminating dominance resistance solutions. For each offspring, we first determine whether it is a dominance resistance solution using Algorithm 5. If it is, no further operation is performed. Otherwise, we compare the dominance relationship between the offspring individual and the parent individual. If they are mutually non-dominating, we compare the change in GaRIF value after replacing the parent individual with the offspring individual. If the GaRIF value does not decrease, the replacement is canceled. The method of selecting the population with the minimum GaRIF value is similar to steps 10–20 in Algorithm 2. At this stage, the population has undergone several iterations and is no longer in the initial chaotic and unordered state. Instead, it exhibits good convergence and diversity. Therefore, steps 3–9 of Algorithm 2 are not required here.

Algorithm 6 GaRIFSelection2

Input: N (Population size), M (Number of objectives), **P** (Population), **F** (Offspring), **HP** (Historical population), h_1 (Control the parameters of gravity), h_2, h_3 (Control the parameters of repulsion)
Output: **P** (Population), E_{GaRIF} (GaRIF of **P**)

```
GM←GaRIFMatrix(N, M, P ∪ F, h2, h3)
for i ← 1 to N do
    DRS ←SearchDRS(N, M, F(i), HP)
    if DRS = true then
        continue
    end if
    if F(i) ≺ P then(i)
        P(i) ← F(i)
    else if P(i) ≺ F then(i)
        continue
    else
        Compute the number of elements in P that dominate P(i) as OldBeDom and the number of elements in P that dominate F(i) as NewBeDom
        if OldBeDom = NewBeDom then
            Calculate the GaRIF values PE and FE for P and F, respectively, using Eqs (5), h1, and GM
            if FE < PE then
                P(i) ← F(i)
            end if
        else if OldBeDom > NewBeDom then
            P(i) ← F(i)
        else
            continue
        end if
    end if
end for
Calculate E_GaRIF of P using Eqs (5), h1, and GM
```

3.5. Perturbation strategy

Most MaOEAs commonly employ simulated binary crossover (SBX) [45] to perform crossover on parents, followed by Gaussian perturbation [46] to mutate the resulting offspring. However, this approach can often lead to a significant proximity between the generated offspring and their parents, thereby restricting the search space and making the population susceptible to getting trapped in local optima. To address this issue, we propose a random perturbation method in Algorithm 7. When the historical minimum of the GaRIF value based on Pareto dominance remains unchanged for a certain number of generations, or when the historical minimum of the sum of distances from each individual in the population to the origin remains unchanged for a certain number of generations in a decomposition-based evolution mechanism, we randomly select a portion of the parent individuals for perturbation and treat the perturbed results as offspring. The perturbation method resembles polynomial mutation (PM) [47], but we replace the Gaussian perturbation in PM with Eqs. (6). The definition of Eqs. (6) is as follows:

$$x_i = x_i \pm \text{random}(lower, upper), \quad i \in (1, D) \tag{6}$$

where $x = \left(x_1, x_2, \ldots, x_D\right)^T$ is an D-dimensional decision vector in the decision space Ω, x_i represents the ith decision variable of the parent individual that has been selected for perturbation, $lower$ represents the lower bound of the decision variable range, while $upper$ represents the upper bound of the decision variable range.

Algorithm 7 VariationOrPerturbation

Input: N (Population size), **P** (Population), V (Perturbation parameter)
Output: **F** (Offspring)

```
for i = 1 → N do
    r1=Random(1)
    if V = 0 || r1 = 1 then
        r2=Random(N)+1
        F(i)' ← Crossover(P(i), P(r2))
        F(i)'' ← Variation(F(i)')
        F← F ∪ F(i)''
    else
        Using Eqs. (6) to replace Gaussian perturbation in polynomial variation to perturb P(i) and generate F(i)
        If the decision variables are outside the valid range, they will be perturbed again.
        F← F ∪ F(i)
    end if
end for
```

Due to the significant randomness in our perturbation, the generated offspring may exhibit substantial differences from their parents, increasing the diversity of the search and aiding in the discovery of potential global optima. To avoid excessive perturbation that could slow down the evolution of the population, we still utilize PM and SBX to generate offspring for the unselected parent individuals.

3.6. Computational complexity

Given a MaOP with M objectives, in the Pareto dominance-based optimization mechanism, the time complexity for non-dominated sorting of a population of size N is $O(N^2M)$, selecting the population with the minimum GaRIF value takes $O(N^2)$ time complexity, and calculating the distance between each pair of individuals in the population requires $O(N^2M)$ time complexity. Therefore, the time complexity for calculating the GaRIF value of the population is $O(N^3M)$, and the overall time complexity of the dominance-based optimization mechanism is $O(N^2M+N^2+N^3M) = O(N^3M)$. In the decomposition-based optimization mechanism, this paper adopts MOEA/D, whose time complexity is $O(MTN)$, where T represents the number of neighboring weight vectors adjacent to a weight vector, and $T < N$. Therefore, the total time complexity of MaOEAIH is $O(N^3M)$.

4. Experimental study

In this section, to verify the effectiveness of MaOEAIH in handling MaOPs, we compared it with several state-of-the-art MaOEAs of different types on a benchmark test suites. These representatives include MaOEAIBP [48], MaOEAIGD [27], ARMOEA [49], HEA [50], tDEA [51], TSNSGAII [52], EFRRR [53], and RVEA [54]. The aforementioned experiments were conducted using the Evolutionary Many-Objective Optimization platform (PlatEMO) [55]. Furthermore, the source code for MaOEAIH and supplementary materials for some experiments are available for access at https://github.com/ASmallChestnut/MaOEAIH.

4.1. Experimental settings

4.1.1. Test problems

In this paper, we employ a set of 31 scalable test problems from widely-used benchmark test suites, namely DTLZ [56], WFG [57],

MAF [58], and IDTLZ [59], to verify the performance of our proposed algorithm on various types of MaOPs. This test suites cover a wide range of MaOP types, including but not limited to unimodal, multimodal, deceptive, hard-to-converge, biased, concave, convex, severely scaled, linear, degenerate, discontinuous, and problems with highly expanded PF. Since the algorithm in this paper focuses on solving different types of MaOPs, MAF14 and MAF15 are primarily used to assess the performance of MaOEAs on problems with large-scale decision variables. Therefore, these two test problems were not adopted in this paper. For the 31 test problems DTLZ1-DTLZ7, WFG1-WFG9, MAF1-MAF13, IDTLZ1-IDTLZ2, we set the number of objectives M as 3, 5, 8, 10, and 15. Except for MAF8 and MAF9, where the decision variable dimensions were set as 2, and MAF13, where the decision variable dimensions were set as 5, the decision variable dimensions for the remaining test problems were set as $D = M + r - 1$. In DTLZ1 and IDTLZ1, r was set as 5, in DTLZ7, r was set as 20, and for the remaining test problems, r was set as 10.

4.1.2. Performance indicators

In this paper, we employ IGD as an evaluation Indicator to assess the overall performance of different algorithms on the aforementioned test suites. IGD provides an approximate assessment of the convergence and diversity of the final solutions by calculating the average distance between each reference point on the true PF and the nearest solution on the approximated PF obtained by the MaOEA. A smaller value of IGD indicates a better overall performance of an algorithm, i.e., a better convergence and diversity. The mathematical expression for IGD is as follows:

$$\mathrm{IGD} = \frac{\sum_{p \in P_{true}} \min_{q \in P_{known}} \|p - q\|}{|P_{true}|} \tag{7}$$

where P_{true} represents the set of reference points obtained from the true PF, P_{known} represents the final solution set obtained by the MaOEA, p denotes a solution from P_{true}, q denotes a solution from P_{known}, $\|p - q\|$ represents the Euclidean distance between p and q, and $|P_{true}|$ represents the number of reference points on the true PF.

Additionally, this paper also employed the DeltaP [60], DM [61], PD [62], and Spread [61] performance Indicators to comprehensively evaluate MaOEAIH's performance in handling various MaOPs. However, given the constraints of space in the paper, these experimental results will be uploaded to the same link as the source code.

4.1.3. Parameter settings

(1) Crossover and Mutation: The crossover operator used by all algorithms in the comparative experiments is SBX, while the mutation operator is PM. For SBX, the crossover probability is set to 1, and the distribution index is set to 20. For PM, the expected number of mutated variables is set to 1, and the distribution index is set to 20.
(2) Population Size: To ensure a fair comparison, the population size in all experiments is uniformly set to 100.
(3) Termination Condition and Number of Runs: In Sections Section 4.2, 4.5, and 4.6, we set the maximum number of function evaluations to 600,000, while in Sections 4.3 and 4.4, it was set to 200,000 and 100,000, respectively. Furthermore, to ensure the reliability of the experimental results, we independently ran each algorithm 30 times on each test problem.
(4) Significance test: We employed the Wilcoxon rank sum [63] test at a significance level of 0.05 to perform statistical analysis on the experimental results. The symbols "+ ", "−", and "≈" indicate that the compared algorithm's results are significantly better than, significantly worse than, or not significantly different from the results of MaOEAIH, respectively. Additionally, the best performing results will be highlighted in dark.

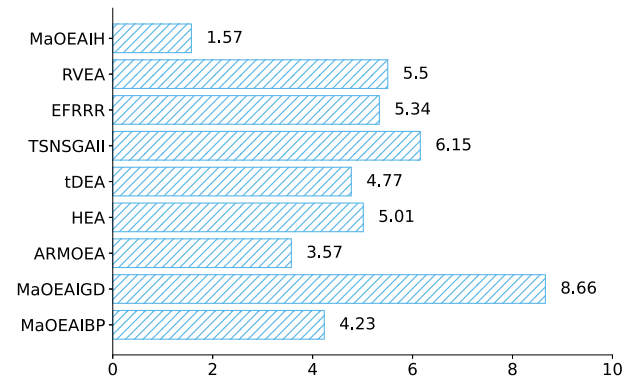

Fig. 8. Average performance score. Smaller scores indicate that the algorithm has better performance on IGD.

4.2. Experimental results and analysis of overall performance of MaOEAIH

The experimental results of all the algorithms in various test suites with different objectives are shown in Tables 2–4. Additionally, we calculated the average performance scores of all algorithms across all test suites, as shown in Fig. 8. Based on these results, it can be observed that MaOEAIH performs the best in terms of IGD values in most of the test problems, exhibiting significant advantages over other competing algorithms. The specific details are as follows:

When dealing with difficult convergence problems such as DTLZ1 and DTLZ3, MaOEAIH adopts a decomposition-based optimization mechanism and a quartile method based on boundary solutions, which effectively assists the population in converging to the PF, thus outperformed other algorithms. It only slightly lags behind EFRRR in the 8-dimensional DTLZ1 problem. To visually demonstrate this result, Fig. 9 presents the parallel coordinate plot of the final populations obtained by each algorithm on the 10-dimensional DTLZ3. From the figure, it is evident that MaOEAIGD, tDEA, and TSNSGAII have objective values exceeding 10 on each objective, failing to converge to the PF. This phenomenon is due to these three algorithms losing selection pressure in high-dimensional spaces, causing individuals to remain distributed far from the PF and unable to approach it. Although ARMOEA, HEA, EFRRR, and RVEA can converge to the PF, their populations' distribution is similar, as they all employ reference points to maintain diversity, resulting in most individuals clustered in certain areas of the PF, with many regions, like the 0.2 to 0.6 objective value range, lacking individual, indicating poor diversity. Similarly, while MaOEAIBP employs a boundary penalty strategy, resulting in a denser distribution in the 0.2 to 0.6 objective value range, it too lacks individual distribution in many PF regions (such as the region with objective values between 0.2 and 0.4), leading to poor diversity. In contrast, MaOEAIH not only converges to the PF but also achieves an even distribution across the entire PF, demonstrating its effective handling of difficult convergence in MaOPs.

In the case of problems like DTLZ2 and DTLZ4, where maintaining population diversity is challenging, MaOEAIH achieves significantly better population diversity compared to other algorithms, thanks to the good performance of GaRIF in preserving population diversity.

In scenarios where problems like DTLZ5, DTLZ6, WFG3, MAF6, MAF8, MAF9, and MAF13 have degenerate PFs, MaOEAIH's results are superior in 65.7% of cases, which can be attributed to GaRIF and the perturbation strategy that enables the population to accurately discover the PF. Fig. 10 displays the population distribution of each algorithm on the 3-dimensional MAF6, and it can be observed that MaOEAIH achieves a more uniform and extensive population distribution, while EFRRR and RVEA fail to converge to the PF due to a lack of intersection between most weight vectors and the PF. Although ARMOEA has a similar population distribution to MaOEAIH, careful observation reveals

Table 2
Mean and standard deviation of the IGD values obtained by MaOEAIH and other MaOEAs for DTLZ test suit.

Problem	M	MaOEAIBP	MaOEAIGD	ARMOEA	HEA	tDEA	TSNSGAII	EFRRR	RVEA	MaOEAIH
DTLZ1	3	2.3002e−2	5.8918e−1	2.0557e−2	2.0558e−2	2.0557e−2	2.9228e−2	2.0557e−2	2.0556e−2	2.0214e−2
		(4.20e−4) −	(3.64e−1) −	(3.94e−7) −	(6.20e−6) −	(6.57e−7) −	(1.40e−2) −	(1.81e−7) −	(8.92e−8) −	(1.32e−4)
	5	7.9549e−2	7.5436e−1	6.8059e−2	6.8060e−2	6.8060e−2	8.5187e−2	6.8058e−2	6.8059e−2	6.6017e−2
		(1.58e−3) −	(6.00e−1) −	(1.28e−6) −	(2.73e−6) −	(1.42e−6) −	(1.56e−2) −	(1.27e−6) −	(4.73e−7) −	(1.20e−3)
	8	1.4086e−1	6.9820e−1	1.0868e−1	1.0870e−1	1.0869e−1	1.6198e−1	1.0858e−1	1.0870e−1	1.0977e−1
		(2.55e−3) −	(9.99e−1) −	(9.35e−6) +	(1.49e−6) +	(1.47e−6) +	(4.29e−2) −	(3.50e−5) +	(1.32e−4) +	(1.91e−3)
	10	1.7945e−1	4.6919e−1	1.5414e−1	1.5416e−1	1.5406e−1	2.3380e−1	1.6026e−1	1.5411e−1	1.2806e−1
		(5.39e−3) −	(2.57e−1) −	(1.82e−4) −	(1.19e−5) −	(4.81e−5) −	(3.97e−2) −	(2.58e−3) −	(6.28e−5) −	(2.84e−3)
	15	1.9927e−1	8.1833e−1	1.7104e−1	1.9532e−1	3.1328e−1	3.8975e−1	2.9480e−1	1.8884e−1	1.5804e−1
		(1.76e−2) −	(6.35e−1) −	(1.40e−2) −	(3.53e−2) −	(5.30e−2) −	(5.14e−2) −	(3.99e−2) −	(5.11e−5) −	(8.18e−3)
DTLZ2	3	7.6659e−2	1.7095e−1	5.4464e−2	5.5213e−2	5.4465e−2	5.4600e−2	5.4465e−2	5.4464e−2	5.2470e−2
		(2.75e−3) −	(1.70e−4) −	(4.30e−7) −	(4.15e−4) −	(5.35e−6) −	(6.98e−5) −	(3.73e−7) −	(2.44e−7) −	(3.25e−4)
	5	2.3207e−1	2.2244e−1	2.1222e−1	2.1222e−1	2.1222e−1	2.1261e−1	2.1221e−1	2.1221e−1	1.9912e−1
		(1.82e−3) −	(2.70e−3) −	(1.24e−5) −	(4.49e−7) −	(1.01e−5) −	(2.35e−4) −	(2.12e−6) −	(2.41e−7) −	(1.09e−3)
	8	4.0331e−1	4.0810e−1	3.8691e−1	3.8698e−1	3.8691e−1	3.8745e−1	3.8680e−1	3.8694e−1	3.7412e−1
		(5.44e−3) −	(1.56e−2) −	(2.74e−5) −	7.10e−4) −	(2.24e−5) −	(1.06e−3) −	(2.36e−4) −	(4.35e−6) −	(1.85e−3)
	10	5.1565e−1	5.9496e−1	4.9214e−1	5.3022e−1	5.0025e−1	5.2575e−1	5.1279e−1	5.0027e−1	4.6382e−1
		(4.63e−3) −	(7.87e−2) −	(6.81e−3) −	(4.21e−3) −	(6.26e−5) −	(4.41e−3) −	(2.25e−3) −	(3.79e−5) −	(1.47e−3)
	15	6.3427e−1	8.9954e−1	6.8944e−1	7.1601e−1	7.0343e−1	7.7475e−1	7.0959e−1	6.9665e−1	5.9439e−1
		(2.30e−3) −	(2.25e−1) −	(1.20e−2) −	(4.94e−3) −	(1.70e−2) −	(7.50e−3) −	(1.25e−2) −	(2.04e−3) −	(1.72e−3)
DTLZ3	3	1.0680e−1	1.4055e+1	5.4467e−2	5.5161e−2	5.4466e−2	7.2548e−2	5.4467e−2	5.4466e−2	5.2627e−2
		(1.71e−1) −	(7.17e+0) −	(5.83e−6) −	(4.47e−4) −	(2.09e−6) −	(4.31e−2) −	(2.71e−6) −	(3.01e−6) −	(5.09e−4)
	5	2.3362e−1	1.2930e+1	2.1223e−1	2.1222e−1	2.1223e−1	2.8723e−1	2.1223e−1	2.1223e−1	2.0135e−1
		(2.27e−3) −	(5.89e+0) −	(2.13e−5) −	(8.42e−6) −	(1.22e−5) −	(3.91e−2) −	(2.01e−5) −	(1.29e−5) −	(2.42e−3)
	8	4.0724e−1	7.9158e+0	3.8780e−1	3.8654e−1	3.9726e−1	6.6207e−1	3.9525e−1	3.8700e−1	3.7992e−1
		(7.01e−3) −	(7.00e+0) −	(1.34e−3) −	(6.17e−4) −	(5.62e−2) −	(1.18e−1) −	(4.48e−2) −	(7.10e−5) −	(1.70e−2)
	10	5.2073e−1	5.8299e+0	5.2648e−1	5.2736e−1	5.2308e−1	1.0323e+0	5.1477e−1	5.0028e−1	4.7096e−1
		(6.97e−3) −	(3.40e+0) −	(1.47e−2) −	(3.88e−3) −	(6.96e−2) −	(7.73e−2) −	(2.07e−3) −	(6.38e−5) −	(7.82e−3)
	15	6.4530e−1	5.6688e+0	7.1025e−1	7.4733e−1	9.4108e−1	1.2509e+2	9.4658e−1	6.9364e−1	6.0326e−1
		(6.74e−3) −	(4.61e+0) −	(3.02e−2) −	(1.15e−1) −	(5.66e−2) −	(6.72e+1) −	(4.81e−2) −	(6.90e−3) −	(9.66e−3)
DTLZ4	3	2.5758e−1	2.6170e−1	3.2737e−1	2.0117e−1	1.0049e−1	5.4639e−2	5.4465e−2	5.4464e−2	5.2850e−2
		(2.79e−1) −	(1.69e−1) −	(2.70e−1) −	(2.27e−1) −	(1.83e−1) −	(6.83e−5) −	(5.91e−7) −	(8.14e−8) −	(5.48e−4)
	5	3.1876e−1	3.5262e−1	2.3978e−1	4.0124e−1	2.1221e−1	2.1264e−1	2.1224e−1	2.2668e−1	2.0216e−1
		(1.12e−1) −	(1.89e−1) −	(7.14e−2) −	(2.56e−1) −	(8.39e−6) −	(1.12e−4) −	(1.30e−4) −	(7.92e−2) −	(1.68e−3)
	8	4.2631e−1	5.0040e−1	3.8691e−1	4.1862e−1	3.8690e−1	3.8727e−1	3.8757e−1	4.2918e−1	3.7725e−1
		(4.74e−2) −	(1.06e−1) −	(4.10e−5) −	(7.33e−2) −	(4.67e−5) −	(4.77e−4) −	(6.70e−4) −	(5.85e−2) −	(1.80e−3)
	10	5.2890e−1	5.9452e−1	5.0178e−1	5.8088e−1	5.0023e−1	5.4773e−1	5.1944e−1	5.3510e−1	4.6783e−1
		(4.18e−2) −	(8.89e−2) −	(3.69e−3) −	(5.62e−2) −	(9.66e−5) −	(4.10e−3) −	(3.37e−3) −	(5.33e−2) −	(1.32e−3)
	15	6.3760e−1	8.3126e−1	6.8935e−1	7.9229e−1	6.9624e−1	1.1736e+0	7.1215e−1	7.6129e−1	5.9563e−1
		(9.81e−3) −	(5.19e−2) −	(9.57e−3) −	(8.17e−2) −	(5.59e−5) −	(2.71e−1) −	(1.37e−2) −	(4.85e−2) −	(1.84e−3)
DTLZ5	3	1.0147e−2	5.6065e−1	5.3590e−3	3.3193e−2	3.2433e−2	1.9252e−2	3.6381e−2	6.2392e−2	4.2210e−3
		(1.32e−3) −	(2.00e−1) −	(1.29e−4) −	(1.88e−3) −	(2.38e−3) −	(4.81e−3) −	(3.78e−3) −	(9.75e−4) −	(8.02e−5)
	5	6.8633e−2	5.7666e−1	7.3456e−2	1.0785e−1	1.8171e−1	1.7016e−1	2.2562e−1	2.2250e−1	5.8045e−2
		(1.27e−2) −	(1.57e−1) −	(1.06e−2) −	(1.93e−2) −	(6.83e−2) −	(5.96e−2) −	(4.92e−2) −	(6.48e−3) −	(1.30e−2)
	8	3.0729e−1	6.4907e−1	1.0610e−1	2.2056e−1	1.9243e−1	6.4882e−1	2.3133e−1	6.7860e−1	6.1577e−2
		(6.24e−2) −	(1.35e−1) −	(2.67e−2) −	(4.43e−2) −	(7.52e−2) −	(8.73e−2) −	(7.31e−2) −	(7.24e−2) −	(4.07e−2)
	10	3.4242e−1	6.5382e−1	1.1405e−1	2.0536e−1	3.1893e−1	6.2932e−1	1.8935e−1	2.2252e−1	5.2282e−2
		(6.99e−7) −	(1.34e−1) −	(2.50e−2) −	(2.92e−2) −	(6.87e−2) −	(8.58e−2) −	(4.14e−2) −	(5.53e−2) −	(2.45e−2)
	15	3.4242e−1	7.1294e−1	1.1649e−1	3.3246e−1	2.4104e−1	4.7108e−1	3.2320e−1	5.5025e−1	4.8561e−2
		(9.36e−7) −	(1.55e−5) −	(2.59e−2) −	(6.75e−2) −	(7.79e−2) −	(2.60e−1) −	(8.82e−2) −	(2.37e−1) −	(2.60e−2)
DTLZ6	3	9.7762e−3	6.7590e−1	5.0647e−3	3.9913e−2	4.6850e−2	3.4961e−2	4.7657e−2	1.1437e−1	4.1203e−3
		(1.28e−3) −	(1.92e−1) −	(6.91e−5) −	(8.48e−4) −	(1.18e−2) −	(2.64e−3) −	(8.83e−3) −	(1.55e−2) −	(3.94e−5)
	5	8.7059e−2	7.1050e−1	7.6135e−2	1.6668e−1	2.9094e−1	3.7612e−1	2.5009e−1	2.3351e−1	9.9003e−2
		(1.37e−2) ≈	(2.14e−1) −	(1.19e−2) ≈	(3.98e−2) −	(8.85e−2) −	(8.64e−2) −	(9.47e−2) −	(4.63e−2) −	(7.20e−2)
	8	3.3750e−1	8.5039e−1	1.1548e−1	3.9543e−1	3.8481e−1	7.2314e−1	2.1736e−1	2.7571e−1	4.3107e−2
		(2.68e−2) −	(3.47e−1) −	(2.69e−2) −	(1.19e−1) −	(9.48e−2) −	(6.47e−2) −	(6.97e−2) −	(8.12e−2) −	(1.67e−2)
	10	3.4242e−1	7.4058e−1	9.7603e−2	4.2196e−1	3.7142e−1	6.2729e−1	2.0585e−1	1.9893e−1	4.3281e−2
		(2.76e−7) −	(1.64e−1) −	(2.58e−2) −	(1.15e−1) −	(8.77e−2) −	(1.07e−1) −	(4.29e−2) −	(5.54e−2) −	(1.81e−2)
	15	3.4242e−1	7.1829e−1	1.1200e−1	6.4578e−1	3.5220e−1	6.3851e+0	2.8090e−1	4.0027e−1	3.9805e−2
		(3.27e−7) −	(1.34e−2) −	(2.47e−2) −	(8.27e−2) −	(1.17e−1) −	(3.10e+0) −	(7.53e−2) −	(1.99e−1) −	(1.92e−2)
DTLZ7	3	1.1639e−1	1.0551e+0	1.7630e−1	2.5084e−1	1.2733e−1	1.0330e−1	1.0629e−1	1.0568e−1	5.8012e−2
		(1.20e−1) ≈	(5.37e−1) −	(1.75e−1) −	(2.06e−1) −	(9.33e−2) −	(6.55e−2) −	(8.07e−3) −	(8.04e−4) −	(1.17e−3)
	5	3.7999e−1	2.0851e+0	3.5130e−1	3.6587e−1	5.3043e−1	3.5486e−1	5.1745e−1	5.0890e−1	2.8988e−1
		(1.25e−1) ≈	(1.12e+0) −	(7.86e−3) −	(1.06e−1) −	(2.51e−2) −	(5.96e−3) −	(3.11e−2) −	(2.00e−3) −	(7.41e−3)
	8	8.7655e−1	3.6858e+0	1.7965e+0	1.1093e+0	1.4518e+0	1.2068e+0	2.1489e+0	1.8555e+0	6.8905e−1
		(1.28e−1) −	(2.02e+0) −	(9.54e−2) −	(3.01e−3) −	(2.12e−1) −	(1.92e−2) −	(4.73e−1) −	(2.71e−2) −	(4.70e−3)
	10	1.3023e+0	4.4274e+0	3.4997e+0	2.1200e+0	1.6480e+0	2.5004e+0	2.0873e+0	3.6403e+0	9.4470e−1
		(3.23e−1) −	(2.42e+0) −	(1.22e−1) −	(1.27e−2) −	(3.07e−1) −	(3.62e−1) −	(5.37e−1) −	(8.75e−2) −	(9.40e−3)
	15	3.3876e+0	2.8895e+0	7.3830e+0	8.2001e+0	1.1100e+1	8.9713e+0	7.9929e+0	7.0858e+0	1.5673e+0
		(7.82e−1) −	(3.45e−1) −	(2.95e−1) −	(1.17e+0) −	(5.45e−1) −	(7.69e−1) −	(5.57e−1) −	(3.08e−1) −	(1.25e−2)
+/−/≈		0/32/3	0/35/0	1/33/1	1/34/0	1/34/0	0/35/0	1/34/0	1/34/0	

a lack of individuals in the central region of its PF. The remaining algorithms fail to completely cover the PF.

In the case of problems with disconnected PFs like DTLZ7 and MAF7, MaOEAIH achieves all the optimal results. This is because

Table 3
Mean and standard deviation of the IGD values obtained by MaOEAIH and other MaOEAs for WFG test suit.

Problem	M	MaOEAIBP	MaOEAIGD	ARMOEA	HEA	tDEA	TSNSGAII	EFRRR	RVEA	MaOEAIH
WFG1	3	1.6193e−1	2.2889e+0	1.4413e−1	1.4501e−1	1.3939e−1	1.5679e−1	1.4209e−1	1.4769e−1	1.6019e−1
		(4.11e−3) −	(5.86e−2) −	(2.90e−3) +	(1.65e−3) +	(1.89e−3) +	(6.43e−3) ≈	(1.62e−3) +	(3.85e−3) +	(9.75e−3)
	5	6.2547e−1	3.3256e+0	4.7412e−1	4.6317e−1	4.4298e−1	5.5745e−1	4.7704e−1	4.4786e−1	4.8527e−1
		(2.37e−2) −	(6.60e−1) −	(3.39e−3) ≈	(4.54e−3) +	(4.05e−3) +	(2.43e−2) −	(1.34e−3) ≈	(9.79e−3) +	(2.61e−2)
	8	1.3562e+0	6.5126e+0	1.0263e+0	1.6722e+0	1.0103e+0	1.4171e+0	9.9490e−1	1.2372e+0	9.7039e−1
		(6.24e−2) −	(2.90e+0) −	(8.78e−3) −	(1.05e−1) −	(3.08e−2) −	(7.47e−2) −	(4.14e−2) −	(1.51e−1) −	(7.27e−2)
	10	1.8482e+0	8.7759e+0	1.2866e+0	1.8211e+0	1.2908e+0	1.8517e+0	1.4422e+0	1.3652e+0	1.1551e+0
		(1.01e−1) −	(4.18e+0) −	(6.45e−3) −	(2.79e−1) −	(4.81e−2) −	(8.23e−2) −	(9.15e−2) −	(1.04e−1) −	(1.08e−1)
	15	2.7299e+0	1.4265e+1	2.0345e+0	3.5698e+0	2.1642e+0	3.2245e+0	2.1556e+0	2.2719e+0	1.8531e+0
		(1.84e−1) −	(5.48e+0) −	(1.60e−2) −	(3.56e−1) −	(8.82e−2) −	(1.78e−1) −	(1.12e−1) −	(6.85e−2) −	(2.24e−1)
WFG2	3	2.2818e−1	1.5063e+0	1.6514e−1	1.5562e−1	1.5829e−1	1.7406e−1	1.7102e−1	1.7397e−1	1.6935e−1
		(1.23e−2) −	(1.71e−1) −	(9.65e−4) +	(1.78e−3) +	(1.09e−3) +	(5.93e−3) −	(2.64e−3) −	(2.21e−3) −	(4.53e−3)
	5	6.2982e−1	1.9751e+0	5.0745e−1	4.9581e−1	4.7278e−1	5.8217e−1	5.1887e−1	4.7875e−1	5.4737e−1
		(1.62e−2) −	(4.48e−1) −	(1.90e−3) +	(8.45e−3) +	(9.01e−4) +	(2.06e−2) −	(1.32e−2) +	(6.66e−3) +	(2.53e−2)
	8	1.3590e+0	2.9085e+0	1.0883e+0	1.8655e+0	1.8210e+0	1.3896e+0	1.1242e+0	1.2923e+0	1.0628e+0
		(3.14e−2) −	(6.77e−1) −	(1.49e−2) −	(3.15e−1) −	(4.34e−1) −	(7.90e−2) −	(5.92e−2) −	(3.14e−2) −	(3.56e−2)
	10	1.9126e+0	5.3011e+0	1.3547e+0	2.1929e+0	3.3610e+0	1.9081e+0	1.4514e+0	1.5087e+0	1.3360e+0
		(8.28e−2) −	(3.06e+0) −	(1.50e−2) ≈	(1.85e−1) −	(5.00e−1) −	(7.27e−2) −	(8.69e−2) −	(1.22e−2) −	(7.83e−2)
	15	2.2712e+0	8.2560e+0	2.0633e+0	3.4293e+0	9.2874e+0	3.4713e+0	3.9870e+0	2.3463e+0	2.1733e+0
		(2.15e−1) ≈	(6.56e+0) −	(3.19e−2) ≈	(3.84e−1) −	(2.14e+0) −	(1.42e−1) −	(9.25e−1) −	(9.20e−3) −	(2.57e−1)
WFG3	3	6.4201e−2	3.1817e+0	1.1622e−1	7.1260e−2	1.2970e−1	1.3271e−1	1.2460e−1	2.2287e−1	5.3865e−2
		(5.14e−3) −	(2.62e−2) −	(7.35e−3) −	(7.99e−3) −	(1.80e−2) −	(2.31e−2) −	(9.23e−3) −	(6.59e−3) −	(2.43e−3)
	5	3.8656e−1	5.4261e+0	6.8676e−1	5.2535e−1	6.3846e−1	7.3887e−1	6.4743e−1	7.1141e−1	3.4749e−1
		(6.02e−2) −	(3.46e−2) −	(4.84e−2) −	(3.57e−2) −	(2.76e−2) −	(1.31e−1) −	(4.34e−2) −	(3.34e−2) −	(3.82e−2)
	8	1.0664e+0	8.8757e+0	2.1489e+0	1.8901e+0	1.8506e+0	2.6711e+0	1.3349e+0	2.4152e+0	1.0756e+0
		(1.87e−1) ≈	(5.35e−2) −	(2.19e−1) −	(9.97e−2) −	(5.14e−1) −	(3.54e−1) −	(3.02e−1) −	(2.91e−1) −	(1.45e−1)
	10	1.6166e+0	1.1191e+1	3.3693e+0	2.6049e+0	2.6522e+0	3.2899e+0	2.2716e+0	3.4125e+0	1.6532e+0
		(2.27e−1) ≈	(4.28e−2) −	(1.83e−1) −	(1.60e−1) −	(8.88e−1) −	(4.59e−1) −	(4.56e−1) −	(7.03e−2) −	(2.87e−1)
	15	3.1522e+0	1.6905e+1	6.4406e+0	7.2867e+0	4.3628e+0	9.5269e+0	4.7111e+0	7.7079e+0	4.2414e+0
		(9.23e−1) +	(1.16e−1) −	(9.06e−1) −	(2.33e−1) −	(5.15e−1) ≈	(5.92e−1) −	(7.22e−1) −	(5.29e−1) −	(9.47e−1)
WFG4	3	3.0289e−1	3.9407e+0	2.2087e−1	2.2286e−1	2.2087e−1	2.2120e−1	2.2091e−1	2.2115e−1	2.0439e−1
		(9.74e−3) −	(3.52e−1) −	(5.52e−6) −	(2.17e−3) −	(3.53e−6) −	(2.81e−4) −	(8.05e−6) −	(3.92e−4) −	(9.38e−4)
	5	1.4128e+0	6.2788e+0	1.2250e+0	1.2251e+0	1.2250e+0	1.2246e+0	1.2262e+0	1.2265e+0	1.1131e+0
		(2.58e−2) −	(5.76e−1) −	(5.38e−5) −	(2.38e−4) −	(7.87e−5) −	(1.11e−3) −	(5.41e−4) −	(1.47e−3) −	(3.25e−3)
	8	3.5379e+0	9.8145e+0	3.5275e+0	3.5330e+0	3.5223e+0	3.5439e+0	3.5609e+0	3.5070e+0	3.1575e+0
		(5.03e−2) −	(1.22e+0) −	(4.60e−3) −	(7.99e−3) −	(1.11e−3) −	(1.14e−2) −	(8.29e−3) −	(1.54e−2) −	(1.19e−2)
	10	5.3858e+0	1.3135e+1	5.8687e+0	6.1211e+0	5.8596e+0	6.0045e+0	5.9145e+0	5.8428e+0	4.8125e+0
		(1.15e−1) −	(2.36e+0) −	(9.87e−3) −	(8.34e−2) −	(2.83e−3) −	(7.56e−2) −	(1.02e−2) −	(9.88e−2) −	(1.75e−2)
	15	9.5276e+0	2.5029e+1	1.2041e+1	1.2717e+1	1.2005e+1	1.3242e+1	1.2033e+1	1.2846e+1	9.4634e+0
		(1.28e−1) −	(3.56e+0) −	(5.39e−2) −	(2.86e−1) −	(1.28e−2) −	(8.73e−2) −	(1.77e−1) −	(4.37e−1) −	(4.05e−2)
WFG5	3	3.0568e−1	7.9349e−1	2.2987e−1	2.3104e−1	2.2987e−1	2.2995e−1	2.3038e−1	2.3000e−1	2.1185e−1
		(1.10e−2) −	(5.61e−1) −	(1.53e−6) −	(9.04e−4) −	(2.74e−6) −	(2.11e−4) −	(5.65e−4) −	(1.90e−4) −	(8.05e−4)
	5	1.4110e+0	6.2667e+0	1.2153e+0	1.2153e+0	1.2153e+0	1.2128e+0	1.2260e+0	1.2158e+0	1.0979e+0
		(2.17e−2) −	(1.05e+0) −	(1.28e−5) −	(1.11e−5) −	(6.27e−6) −	(9.91e−4) −	(1.38e−2) −	(5.65e−4) −	(2.30e−3)
	8	3.5491e+0	1.2887e+1	3.5269e+0	3.5353e+0	3.5279e+0	3.5281e+0	3.5736e+0	3.5029e+0	3.1670e+0
		(4.11e−2) −	(2.35e+0) −	(2.40e−3) −	(5.66e−3) −	(6.29e−5) −	(7.00e−3) −	(2.04e−2) −	(9.38e−3) −	(1.03e−2)
	10	5.4190e+0	1.6466e+1	5.8128e+0	6.0313e+0	5.8201e+0	5.8801e+0	6.1714e+0	5.7981e+0	4.8047e+0
		(1.27e−1) −	(2.86e+0) −	(9.76e−3) −	(5.73e−2) −	(3.78e−4) −	(3.89e−2) −	(9.44e−2) −	(2.23e−2) −	(1.67e−2)
	15	9.6080e+0	2.7812e+1	1.1868e+1	1.2225e+1	1.1940e+1	1.3101e+1	1.2031e+1	1.1433e+1	9.4201e+0
		(1.45e−1) −	(2.94e+0) −	(1.08e−1) −	(1.16e−1) −	(4.25e−2) −	(1.67e−1) −	(1.69e−1) −	(2.28e−1) −	(4.92e−2)
WFG6	3	3.1686e−1	2.7652e+0	2.3496e−1	2.3938e−1	2.3985e−1	2.4285e−1	2.3350e−1	2.3507e−1	2.2220e−1
		(1.00e−2) −	(1.32e+0) −	(6.97e−3) −	(1.03e−2) −	(9.96e−3) −	(1.09e−2) −	(9.31e−3) −	(9.90e−3) −	(8.22e−3)
	5	1.4186e+0	5.6896e+0	1.2141e+0	1.2145e+0	1.2143e+0	1.2135e+0	1.2154e+0	1.2151e+0	1.1104e+0
		(2.21e−2) −	(1.28e+0) −	(7.88e−4) −	(1.20e−3) −	(9.53e−4) −	(1.36e−3) −	(3.56e−3) −	(1.24e−3) −	(2.81e−3)
	8	3.6292e+0	9.4138e+0	3.5314e+0	3.5499e+0	3.5331e+0	3.5507e+0	3.5500e+0	3.5917e+0	3.1677e+0
		(4.00e−2) −	(5.08e+0) −	(5.58e−3) −	(9.71e−3) −	(4.85e−3) −	(1.98e−2) −	(5.77e−3) −	(4.60e−2) −	(1.12e−2)
	10	5.7019e+0	8.9150e+0	5.7876e+0	6.2010e+0	5.8043e+0	6.1258e+0	5.8571e+0	6.0978e+0	4.8138e+0
		(7.86e−2) −	(4.77e+0) −	(2.11e−2) −	(1.03e−1) −	(1.26e−2) −	(8.12e−2) −	(2.01e−2) −	(1.71e−1) −	(1.83e−2)
	15	9.7708e+0	2.1076e+1	1.2015e+1	1.2828e+1	1.1914e+1	1.3583e+1	1.1948e+1	1.3026e+1	9.4432e+0
		(2.06e−1) −	(9.56e+0) −	(1.97e−1) −	(2.43e−1) −	(2.24e−2) −	(8.88e−2) −	(1.93e−1) −	(3.72e−1) −	(5.11e−2)
WFG7	3	3.0883e−1	2.5716e+0	2.2090e−1	2.2377e−1	2.2089e−1	2.2210e−1	2.2089e−1	2.2102e−1	2.0772e−1
		(1.09e−2) −	(8.79e−1) −	(2.00e−5) −	(1.87e−3) −	(5.98e−6) −	(6.09e−4) −	(7.17e−6) −	(4.74e−5) −	(1.60e−3)
	5	1.4338e+0	5.8209e+0	1.2264e+0	1.2268e+0	1.2258e+0	1.2236e+0	1.2292e+0	1.2310e+0	1.1227e+0
		(1.93e−2) −	(6.87e−1) −	(6.14e−4) −	(1.39e−3) −	(3.82e−4) −	(1.36e−3) −	(2.82e−3) −	(3.30e−3) −	(4.49e−3)
	8	3.6012e+0	1.0736e+1	3.5337e+0	3.5473e+0	3.5317e+0	3.5536e+0	3.5958e+0	3.7109e+0	3.1924e+0
		(4.61e−2) −	(2.21e+0) −	(7.85e−3) −	(8.32e−3) −	(1.04e−2) −	(2.41e−2) −	(1.01e−2) −	(7.60e−2) −	(1.51e−2)
	10	5.5732e+0	1.3798e+1	5.8842e+0	6.2353e+0	5.8722e+0	6.0813e+0	5.9478e+0	6.1690e+0	4.8348e+0
		(7.52e−2) −	(2.05e+0) −	(5.82e−2) −	(8.07e−2) −	(1.11e−2) −	(5.42e−2) −	(1.30e−2) −	(1.12e−1) −	(2.11e−2)
	15	9.4888e+0	2.4122e+1	1.1706e+1	1.2620e+1	1.1999e+1	1.3522e+1	1.3495e+1	1.2719e+1	9.4874e+0
		(9.16e−2) ≈	(2.35e+0) −	(1.97e−1) −	(1.98e−1) −	(4.97e−2) −	(2.17e−1) −	(1.64e+0) −	(3.35e+0) −	(7.57e−2)

(*continued on next page*)

MaOEAIH uses the populations with PF shape information as the weight vector, effectively avoiding the waste of weight vectors and the issue of uneven coverage of weight vectors on the PF.

In problems with convex PFs such as WFG1, WFG2, MAF3, MAF10, and MAF11, MaOEAIH only obtains approximately half of the optimal results. This may attribute to the failure of quartile method based on

Table 3 (*continued*).

WFG8	3	3.5120e−1	3.6732e+0	2.6394e−1	2.7339e−1	2.7335e−1	3.0022e−1	2.8369e−1	2.8151e−1	2.5585e−1
		(1.16e−2) −	(2.22e−1) −	(2.38e−3) −	(2.79e−3) −	(3.63e−3) −	(5.79e−3) −	(4.60e−3) −	(4.30e−3) −	(3.09e−3)
	5	1.3516e+0	5.4034e+0	1.2219e+0	1.2263e+0	1.2415e+0	1.2312e+0	1.3374e+0	1.2391e+0	1.1468e+0
		(1.82e−2) −	(1.80e+0) −	(8.11e−4) −	(4.21e−3) −	(1.06e−2) −	(2.91e−3) −	(2.08e−2) −	(8.98e−3) −	(4.82e−3)
	8	3.5232e+0	1.2010e+1	3.5985e+0	3.7501e+0	3.5812e+0	3.6797e+0	3.5799e+0	3.9253e+0	3.2796e+0
		(2.89e−2) −	(2.30e+0) −	(2.52e−2) −	(2.45e−2) −	(2.02e−2) −	(1.92e−2) −	(8.73e−2) −	(1.10e−1) −	(2.06e−2)
	10	5.4949e+0	1.4874e+1	5.8950e+0	6.0394e+0	5.7271e+0	6.0662e+0	5.8898e+0	6.0830e+0	4.9222e+0
		(1.30e−1) −	(1.62e+0) −	(1.12e−1) −	(4.94e−2) −	(5.37e−2) −	(7.04e−2) −	(1.31e−1) −	(1.97e−1) −	(7.27e−2)
	15	9.5207e+0	2.7044e+1	1.2005e+1	1.2947e+1	1.2310e+1	1.2947e+1	1.3031e+1	1.2232e+1	9.6363e+0
		(1.97e−1) +	(2.49e+0) −	(3.82e−1) −	(1.10e−1) −	(4.41e−1) −	(1.71e−1) −	(3.77e−1) −	(4.36e−1) −	(1.09e−1)
WFG9	3	2.9857e−1	2.0939e+0	2.2051e−1	2.2183e−1	2.2040e−1	2.2204e−1	2.2989e−1	2.2110e−1	2.0074e−1
		(1.31e−2) −	(5.41e−1) −	(3.46e−4) −	(1.49e−3) −	(2.70e−4) −	(1.02e−3) −	(2.56e−3) −	(7.70e−4) −	(1.10e−3)
	5	1.3648e+0	4.8164e+0	1.2149e+0	1.2154e+0	1.2142e+0	1.2088e+0	1.3605e+0	1.2188e+0	1.1053e+0
		(3.28e−2) −	(1.16e+0) −	(2.98e−3) −	(2.37e−3) −	(1.98e−3) −	(7.61e−3) −	(1.94e−2) −	(2.92e−3) −	(5.05e−3)
	8	3.4308e+0	1.1188e+1	3.5357e+0	3.5501e+0	3.5144e+0	3.5868e+0	3.5581e+0	3.5116e+0	3.2215e+0
		(2.95e−2) −	(3.66e+0) −	(1.21e−2) −	(1.57e−2) −	(1.24e−2) −	(2.52e−2) −	(7.68e−2) −	(2.23e−2) −	(2.34e−2)
	10	5.1881e+0	1.5208e+1	5.8126e+0	5.8813e+0	5.8293e+0	5.9208e+0	6.2385e+0	5.8113e+0	4.8333e+0
		(1.03e−1) −	(3.96e+0) −	(1.26e−2) −	(4.54e−2) −	(3.65e−2) −	(6.67e−2) −	(7.58e−2) −	(6.54e−2) −	(3.98e−2)
	15	9.1811e+0	2.6112e+1	1.1596e+1	1.2039e+1	1.1938e+1	1.2775e+1	1.1632e+1	1.1303e+1	9.5022e+0
		(8.81e−2) +	(6.40e+0) −	(2.78e−1) −	(5.57e−2) −	(2.68e−1) −	(2.35e−1) −	(5.73e−1) −	(4.38e−1) −	(1.06e−1)
+/−/≈		3/38/4	0/45/0	3/39/3	4/41/0	4/40/1	0/44/1	2/42/1	3/42/0	

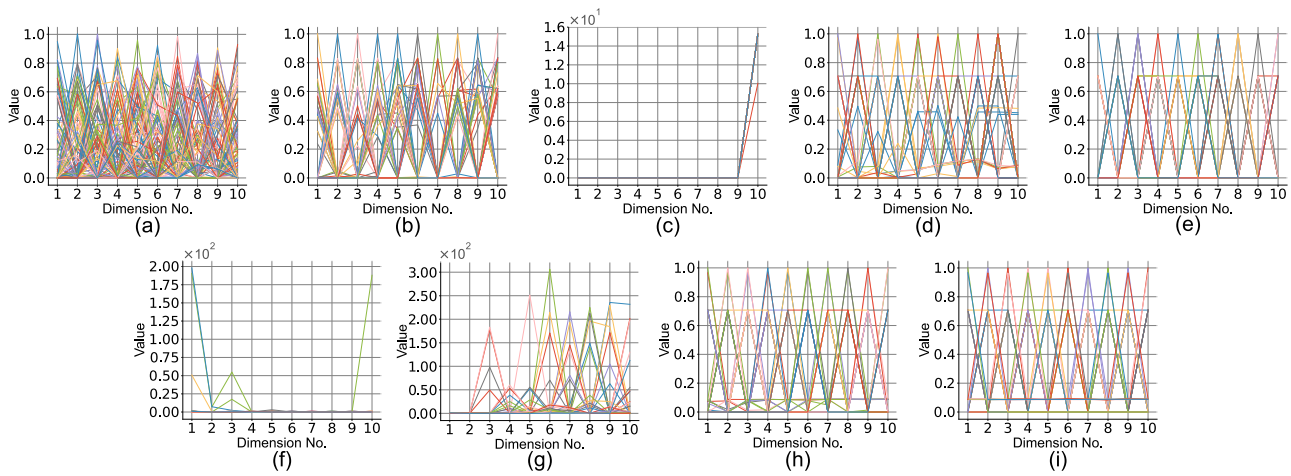

Fig. 9. Distribution of the final population for all algorithms on DTLZ3 problem with 10 objectives. The population obtained by MaOEAIH exhibits superior convergence and diversity compared to other algorithms. (a)MaOEAIH. (b)MaOEAIBP. (c)MaOEAIGD. (d)ARMOEA. (e)HEA. (f)tDEA. (g)TSNSGAII. (h)EFRRR. (i)RVEA.

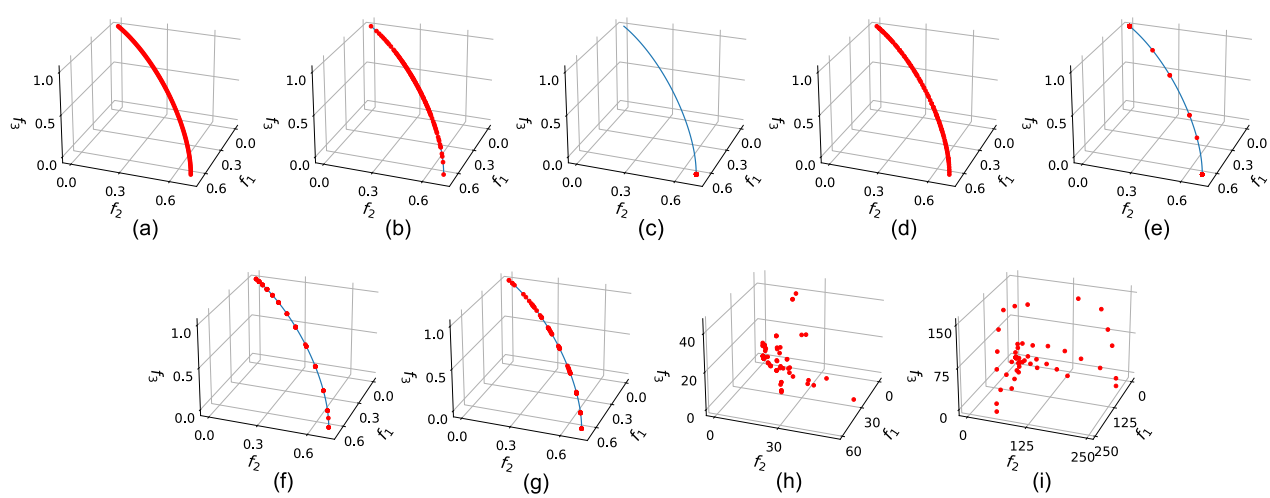

Fig. 10. Distribution of the final population for all algorithms on MaF6 problem with 3 objectives. The population obtained by MaOEAIH exhibits superior coverage and uniformity compared to other algorithms. (a)MaOEAIH. (b)MaOEAIBP. (c)MaOEAIGD. (d)ARMOEA. (e)HEA. (f)tDEA. (g)TSNSGAII. (h)EFRRR. (i)RVEA.

boundary solutions in accurately identifying the range of the PF, resulting in compromised population diversity. Nevertheless, MaOEAIH's results are mostly superior to the other compared algorithms.

For WFG4 to WFG9 and MAF12 with characteristics of multimodality, large basin sizes, deception, separable variables, and nonseparable variables in the search space, except for the 15-dimensional WFG8,

Table 4
Mean and standard deviation of the IGD values obtained by MaOEAIH and other MaOEAs for MAF and IDTLZ test suits.

Problem	M	MaOEAIBP	MaOEAIGD	ARMOEA	HEA	tDEA	TSNSGAII	EFRRR	RVEA	MaOEAIH
MaF1	3	4.4589e−2	1.6193e−1	4.3539e−2	7.0720e−2	8.0826e−2	7.1340e−2	7.7803e−2	8.2142e−2	4.0410e−2
		(7.02e−4) −	(4.61e−4) −	(1.88e−4) −	(2.29e−4) −	(1.10e−3) −	(7.89e−4) −	(3.69e−4) −	(1.28e−4) −	(1.19e−4)
	5	1.4909e−1	2.9193e−1	1.4860e−1	2.2812e−1	5.8844e−1	2.1666e−1	2.5535e−1	4.3712e−1	1.3113e−1
		(2.18e−3) −	(6.66e−3) −	(3.24e−3) −	(5.69e−4) −	(1.75e−1) −	(2.75e−3) −	(2.04e−2) −	(1.32e−1) −	(3.24e−4)
	8	2.6383e−1	3.7260e−1	2.6524e−1	3.4513e−1	3.2485e−1	3.5083e−1	4.5968e−1	6.6904e−1	2.1862e−1
		(4.45e−3) −	(4.01e−3) −	(1.34e−3) −	(6.30e−3) −	(1.29e−2) −	(1.10e−2) −	(7.70e−3) −	(7.73e−2) −	(3.14e−4)
	10	3.2051e−1	3.7113e−1	3.0245e−1	3.4325e−1	3.6306e−1	3.5632e−1	4.7007e−1	7.1447e−1	2.5467e−1
		(7.39e−3) −	(1.85e−2) −	(5.18e−4) −	(5.91e−3) −	(1.65e−2) −	(1.20e−2) −	(1.87e−2) −	(7.94e−2) −	(8.22e−4)
	15	4.0021e−1	4.0315e−1	4.4023e−1	4.1330e−1	3.9415e−1	4.6409e−1	5.8125e−1	7.7056e−1	3.0401e−1
		(1.29e−2) −	(2.36e−2) −	(5.41e−2) −	(1.35e−2) −	(2.10e−2) −	(2.80e−2) −	(2.38e−2) −	(7.80e−2) −	(9.35e−4)
MaF2	3	3.1420e−2	2.7048e−1	3.3238e−2	3.6444e−2	3.6442e−2	4.6275e−2	3.7112e−2	4.0009e−2	2.9211e−2
		(6.16e−4) −	(1.77e−1) −	(7.82e−4) −	(1.08e−4) −	(3.93e−4) −	(2.22e−3) −	(4.15e−4) −	(4.04e−4) −	(3.27e−4)
	5	1.2926e−1	1.4906e−1	1.2189e−1	1.3592e−1	1.4441e−1	1.4811e−1	1.4724e−1	1.4576e−1	1.1828e−1
		(3.80e−3) −	(5.18e−2) −	(1.44e−3) −	(8.44e−4) −	(2.70e−3) −	(1.27e−3) −	(5.33e−3) −	(9.43e−4) −	(1.40e−3)
	8	2.0269e−1	3.9577e−1	2.0042e−1	1.8077e−1	2.0928e−1	1.8381e−1	2.3934e−1	5.0894e−1	1.7210e−1
		(9.85e−3) −	(1.99e−2) −	(4.48e−3) −	(1.64e−3) −	(1.10e−2) −	(8.56e−4) −	(2.01e−2) −	(2.26e−1) −	(2.41e−3)
	10	2.3948e−1	4.1771e−1	2.4284e−1	2.1012e−1	3.0614e−1	2.1506e−1	2.5383e−1	6.5393e−1	1.9873e−1
		(3.11e−2) −	(1.83e−2) −	(1.01e−2) −	(2.19e−3) −	(3.21e−2) −	(1.89e−3) −	(1.89e−2) −	(1.80e−1) −	(2.57e−3)
	15	4.1890e−1	4.2886e−1	3.8133e−1	2.2375e−1	5.3500e−1	2.4045e−1	6.8898e−1	8.4207e−1	2.3376e−1
		(1.44e−2) −	(2.32e−2) −	(3.51e−2) −	(5.34e−4) +	(8.95e−2) −	(8.18e−3) −	(3.62e−2) −	(3.45e−2) −	(4.78e−3)
MaF3	3	4.0278e−2	2.6135e+2	4.6488e−2	3.9784e−2	5.0037e−2	1.4308e−1	4.6358e−2	5.9056e−2	3.1995e−2
		(1.18e−3) −	(2.12e+2) −	(6.26e−5) −	(2.75e−4) −	(4.00e−5) −	(7.62e−2) −	(1.66e−5) −	(6.89e−2) −	(2.71e−4)
	5	9.0986e−2	3.6916e+1	9.8598e−2	8.0785e−2	1.1019e−1	1.7801e−1	9.8640e−2	8.1630e−2	8.3555e−2
		(2.80e−3) −	(5.48e+1) −	(2.42e−4) −	(2.21e−3) +	(1.35e−4) −	(4.69e−2) −	(5.74e−5) −	(5.39e−3) ≈	(2.51e−3)
	8	1.4587e−1	2.8851e+1	1.4150e−1	1.6639e−1	3.1120e−1	5.6779e+5	1.8414e−1	1.1020e−1	1.2158e−1
		(5.93e−3) −	(4.81e+1) −	(8.17e−4) −	(4.59e−3) −	(2.95e−1) −	(2.06e+6) −	(4.29e−2) −	(2.27e−2) +	(1.11e−2)
	10	1.7262e−1	5.0830e+1	1.2118e−1	1.7522e−1	4.6567e−1	3.1849e+6	2.2786e−1	9.9087e−2	1.3119e−1
		(3.49e−2) −	(1.23e+2) −	(1.10e−3) +	(6.77e−3) −	(3.27e−1) −	(1.07e+7) −	(5.98e−2) −	(5.33e−3) +	(9.68e−3)
	15	1.1687e−1	3.3237e+1	1.2708e−1	1.8336e−1	7.9379e−1	2.8111e+8	3.6519e−1	1.2223e−1	1.5026e−1
		(7.68e−3) +	(3.61e+1) −	(5.06e−3) +	(1.31e−2) −	(1.94e−1) −	(1.44e+9) −	(1.54e−1) −	(4.97e−3) +	(1.16e−2)
MaF4	3	3.2256e−1	8.2167e+0	3.3946e−1	3.3922e−1	3.2337e−1	4.1327e−1	3.8067e−1	4.1322e−1	2.4795e−1
		(1.36e−2) −	(7.32e+0) −	(1.80e−3) −	(5.21e−3) −	(1.44e−2) −	(3.62e−2) −	(2.02e−2) −	(9.68e−2) −	(1.32e−2)
	5	2.4349e+0	4.4789e+1	2.9880e+0	2.8725e+0	4.1436e+0	3.4602e+0	4.0326e+0	4.4791e+0	1.9693e+0
		(6.12e−2) −	(3.93e+1) −	(2.17e−1) −	(5.11e−2) −	(5.71e−1) −	(3.97e−1) −	(2.85e−1) −	(1.21e+0) −	(1.18e−2)
	8	1.7440e+1	2.0899e+2	3.0201e+1	3.2914e+1	4.0393e+1	3.9342e+1	4.5481e+1	5.9446e+1	1.6887e+1
		(1.12e+0) −	(2.47e+1) −	(2.02e+0) −	(2.01e+0) −	(6.83e+0) −	(1.70e+0) −	(5.51e+0) −	(1.59e+1) −	(1.16e+0)
	10	6.9016e+1	9.6578e+2	1.4429e+2	1.4009e+2	1.7957e+2	1.4396e+2	1.5426e+2	2.2986e+2	6.7293e+1
		(5.74e+0) ≈	(2.81e+2) −	(1.73e+1) −	(3.63e+0) −	(1.48e+1) −	(3.48e+0) −	(1.67e+1) −	(4.85e+1) −	(3.96e+0)
	15	2.2534e+3	8.4171e+4	7.0742e+3	5.3771e+3	5.6267e+3	5.7033e+3	8.2495e+3	9.5027e+3	2.4019e+3
		(2.66e+2) ≈	(6.24e+4) −	(1.24e+3) −	(3.24e+2) −	(3.50e+2) −	(2.44e+2) −	(1.48e+3) −	(2.50e+3) −	(3.69e+2)
MaF5	3	1.1069e+0	1.5570e+0	1.1028e+0	1.2017e+0	3.1915e−1	2.6041e−1	2.5976e−1	3.0117e−1	2.3366e−1
		(1.22e+0) −	(7.17e−1) −	(1.28e+0) −	(1.45e+0) −	(3.25e−1) −	(6.03e−4) −	(2.32e−6) −	(2.27e−1) −	(2.44e−3)
	5	3.3244e+0	5.9378e+0	2.4997e+0	3.6409e+0	2.3735e+0	2.3758e+0	2.3744e+0	3.0815e+0	2.0173e+0
		(1.41e+0) −	(2.25e+0) −	(6.36e−1) −	(2.01e+0) −	(6.12e−4) −	(6.73e−3) −	(2.05e−3) −	(1.31e+0) −	(1.84e−2)
	8	2.1506e+1	4.2269e+1	2.8482e+1	2.7974e+1	2.8236e+1	2.7799e+1	2.7748e+1	2.9877e+1	1.6225e+1
		(5.13e+0) −	(6.69e+0) −	(3.53e−1) −	(3.29e+0) −	(1.18e−2) −	(4.03e−1) −	(3.84e−1) −	(3.81e+0) −	(5.46e−1)
	10	8.4867e+1	2.9783e+2	1.6678e+2	1.2346e+2	1.3717e+2	1.4085e+2	1.4021e+2	1.3191e+2	6.7021e+1
		(1.08e+1) −	(3.70e+1) −	(7.02e+0) −	(1.62e+1) −	(1.63e−1) −	(3.70e+0) −	(1.67e+0) −	(1.70e+1) −	(9.68e+0)
	15	2.4997e+3	7.3234e+3	5.8753e+3	3.6731e+3	4.8127e+3	5.3679e+3	4.9498e+3	6.6090e+3	1.9825e+3
		(3.78e+2) −	(1.02e+1) −	(2.72e+2) −	(4.74e+2) −	(2.10e+0) −	(5.18e+2) −	(7.16e+1) −	(1.02e+3) −	(1.30e+2)
MaF6	3	9.6296e−3	6.7058e−1	5.1340e−3	3.9256e−2	3.6147e−2	3.3861e−2	3.7227e−2	3.5524e−2	4.0351e−3
		(9.80e−4) −	(1.05e−1) −	(1.33e−4) −	(1.32e−3) −	(5.04e−3) −	(4.05e−3) −	(1.10e−2) −	(1.42e−3) −	(2.98e−5)
	5	9.8685e−3	6.8324e−1	5.0810e−3	7.9791e−2	1.4659e−1	6.5538e−2	1.0936e−1	8.1554e−2	4.0852e−3
		(1.13e−3) −	(3.92e−2) −	(6.43e−5) −	(3.61e−3) −	(2.11e−2) −	(1.01e−2) −	(2.41e−2) −	(1.30e−2) −	(4.26e−5)
	8	7.6379e−2	7.0808e−1	6.3019e−3	1.2281e−1	1.5180e−1	1.2153e−1	1.6648e−1	5.1535e−1	4.4796e−3
		(1.35e−1) −	(2.64e−2) −	(3.27e−4) −	(1.77e−2) −	(6.94e−2) −	(1.02e−2) −	(3.04e−2) −	(2.76e−1) −	(1.72e−3)
	10	1.6526e−1	7.1319e−1	9.4521e−3	1.3389e−1	2.3817e−1	2.6595e−1	1.8880e−1	1.3144e−1	3.8410e−2
		(1.69e−1) −	(2.08e−2) −	(4.58e−3) ≈	(3.27e−2) −	(7.35e−2) −	(1.45e−1) −	(3.57e−2) −	(4.26e−4) −	(7.24e−2)
	15	3.0914e−1	7.1299e−1	4.8725e−2	2.7436e−1	2.6753e−1	3.2419e−1	3.5369e−1	5.9375e−1	9.4281e−2
		(1.01e−1) −	(1.44e−4) −	(2.19e−2) ≈	(5.82e−2) −	(7.32e−2) −	(3.88e−2) −	(6.42e−2) −	(2.03e−1) −	(9.07e−2)
MaF7	3	1.0716e−1	9.0481e−1	1.7478e−1	1.5811e−1	1.2992e−1	1.0379e−1	1.0695e−1	1.0532e−1	5.7637e−2
		(1.12e−1) ≈	(5.05e−1) −	(2.55e−1) −	(1.25e−1) −	(9.26e−2) −	(6.57e−2) −	(6.70e−3) −	(2.68e−4) −	(1.19e−3)
	5	3.5717e−1	1.9242e+0	3.4645e−1	5.8827e−1	5.2898e−1	3.5593e−1	5.1688e−1	5.0918e−1	2.8967e−1
		(1.19e−1) ≈	(1.15e+0) −	(5.48e−3) −	(5.25e−1) −	(2.48e−2) −	(5.67e−3) −	(2.62e−2) −	(4.06e−4) −	(7.14e−3)
	8	8.2739e−1	3.8206e+0	1.8084e+0	1.1242e+0	1.4042e+0	1.2123e+0	2.0829e+0	1.8280e+0	6.8838e−1
		(9.91e−2) −	(1.99e+0) −	(5.17e−2) −	(7.83e−2) −	(2.52e−1) −	(2.21e−2) −	(4.77e−1) −	(1.41e−1) −	(3.61e−3)
	10	1.2774e+0	4.4074e+0	3.4751e+0	2.1207e+0	1.7420e+0	2.6453e+0	1.9900e+0	3.6522e+0	9.4205e−1
		(2.90e−1) −	(2.45e+0) −	(1.23e−1) −	(1.23e−2) −	(4.31e−1) −	(3.66e−1) −	(4.10e−1) −	(2.48e−2) −	(9.65e−3)
	15	3.2246e+0	3.1876e+0	7.3758e+0	8.0627e+0	1.1100e+1	8.9916e+0	8.1645e+0	7.1575e+0	1.5692e+0
		(8.03e−1) −	(1.70e+0) −	(3.24e−1) −	(8.23e−1) −	(5.49e−1) −	(5.99e−1) −	(4.66e−1) −	(1.73e−1) −	(1.29e−2)

(*continued on next page*)

WFG9, and MAF12, MaOEAIH achieves all the optimal results. To showcase this outstanding result, Fig. 11 presents the parallel coordinate plot of the final populations obtained by each algorithm on the 10-dimensional WFG5. From the figure, it can be seen that MaOEAIH exhibits a wider and more uniformly distributed region compared to other algorithms. Although the population distribution obtained by

Table 4 (*continued*).

MaF8	3	6.5136e−2	5.7543e−1	7.6668e−2	2.2909e−1	1.9392e−1	1.4783e−1	1.3977e−1	1.4060e−1	6.8220e−2
		(1.37e−3) +	(2.21e−1) −	(2.29e−3) −	(3.40e−1) −	(5.80e−2) −	(1.26e−2) −	(1.67e−2) −	(7.85e−3) −	(1.32e−3)
	5	1.1554e−1	7.4477e−1	1.3856e−1	2.6330e−1	4.1522e−1	2.4935e−1	3.6984e−1	4.6734e−1	1.1122e−1
		(2.63e−3) −	(1.56e−1) −	(5.35e−3) −	(2.79e−1) −	(2.10e−2) −	(2.11e−2) −	(1.21e−2) −	(5.08e−2) −	(1.50e−3)
	8	1.5862e−1	1.3575e+0	2.1529e−1	2.7587e−1	7.1531e−1	2.7966e−1	9.0911e−1	9.7908e−1	1.4661e−1
		(3.88e−3) −	(2.98e−1) −	(1.01e−2) −	(2.35e−3) −	(1.01e−1) −	(1.47e−2) −	(8.66e−2) −	(1.09e−1) −	(1.48e−3)
	10	1.7711e−1	1.4657e+0	2.4192e−1	4.0508e−1	1.0347e+0	4.5049e−1	8.7934e−1	1.1321e+0	1.6333e−1
		(3.83e−3) −	(2.22e−1) −	(5.78e−3) −	(2.78e−2) −	(1.47e−1) −	(5.30e−2) −	(1.66e−1) −	(1.37e−1) −	(1.51e−3)
	15	2.1556e−1	2.0019e+0	5.0385e−1	5.6157e−1	1.4715e+0	5.9553e−1	1.2557e+0	1.5852e+0	1.9907e−1
		(2.73e−3) −	(2.60e−1) −	(3.61e−2) −	(5.58e−2) −	(1.77e−1) −	(5.38e−2) −	(1.65e−1) −	(2.16e−1) −	(1.40e−3)
MaF9	3	6.9543e−2	7.8781e−1	6.2000e−2	6.1988e−2	6.1992e−2	4.2021e−1	6.6023e−2	6.1989e−2	6.0692e−2
		(1.70e−3) −	(3.55e−1) −	(6.87e−5) −	(9.42e−6) −	(9.09e−7) −	(5.71e−2) −	(5.74e−4) −	(2.33e−6) −	(3.71e−4)
	5	1.1501e−1	9.2055e−1	1.4580e−1	1.5849e−1	8.1010e−1	6.1751e−1	3.8870e−1	4.2101e−1	1.0781e−1
		(1.77e−3) −	(4.21e−1) −	(7.93e−3) −	(1.34e−2) −	(2.22e−1) −	(1.93e−1) −	(5.96e−2) −	(9.85e−2) −	(3.67e−3)
	8	1.5656e−1	1.2707e+0	2.5974e−1	2.2230e−1	1.2268e+0	2.3992e+0	5.4955e−1	9.9538e−1	1.0777e+0
		(1.90e−3) +	(5.69e−1) ≈	(1.24e−2) +	(2.16e−2) +	(8.88e−1) ≈	(2.00e+0) −	(8.11e−2) +	(2.29e−1) ≈	(3.87e−1)
	10	1.8424e−1	1.7338e+0	3.2110e−1	4.6984e−1	1.0978e+0	1.2997e+0	7.3699e−1	1.3284e+0	1.8774e+0
		(3.03e−3) +	(9.56e−1) ≈	(2.44e−2) +	(1.81e−2) +	(1.24e−1) +	(3.65e−1) +	(1.08e−1) +	(4.13e−1) +	(8.67e−1)
	15	2.2086e−1	7.0181e+0	5.1097e−1	1.1867e+0	1.8377e+0	1.8914e+0	1.3581e+0	2.9038e+0	2.2225e−1
		(3.24e−3) +	(5.39e+0) −	(3.82e−2) −	(2.21e−1) −	(5.11e−1) −	(2.10e+0) −	(2.11e−1) −	(3.17e+0) −	(5.23e−2)
MaF10	3	1.6083e−1	2.2668e+0	1.4481e−1	1.4533e−1	1.3968e−1	1.5625e−1	1.4183e−1	1.4543e−1	1.6109e−1
		(5.83e−3) ≈	(1.33e−1) −	(2.73e−3) +	(1.59e−3) +	(1.52e−3) +	(8.00e−3) ≈	(1.49e−3) +	(3.31e−3) +	(1.30e−2)
	5	6.3411e−1	3.3187e+0	4.7436e−1	4.6253e−1	4.4383e−1	5.5099e−1	4.7681e−1	4.4681e−1	4.9232e−1
		(2.22e−2) −	(6.97e−1) −	(4.28e−3) +	(3.96e−3) +	(6.51e−3) +	(1.81e−2) −	(1.51e−3) +	(4.88e−3) +	(2.93e−2)
	8	1.3554e+0	7.0777e+0	1.0190e+0	1.6900e+0	1.0099e+0	1.4458e+0	9.9164e−1	1.2782e+0	9.6382e−1
		(6.54e−2) −	(2.38e+0) −	(8.65e−3) −	(1.49e−1) −	(5.90e−2) −	(7.46e−2) −	(3.35e−2) −	(1.46e−1) −	(6.04e−2)
	10	1.8399e+0	7.4322e+0	1.2879e+0	1.8072e+0	1.2771e+0	1.8886e+0	1.4397e+0	1.3768e+0	1.1363e+0
		(8.68e−2) −	(3.32e+0) −	(8.26e−3) −	(2.99e−1) −	(2.91e−2) −	(1.28e−1) −	(1.08e−1) −	(8.33e−2) −	(7.75e−2)
	15	2.7222e+0	1.2930e+1	2.0356e+0	3.5118e+0	2.2123e+0	3.2306e+0	2.1757e+0	2.2723e+0	1.8487e+0
		(2.04e−1) −	(6.94e+0) −	(1.38e−2) −	(3.96e−1) −	(1.42e−1) −	(1.95e−1) −	(1.33e−1) −	(5.80e−2) −	(2.17e−1)
MaF11	3	2.2847e−1	1.5486e+0	1.6505e−1	1.5604e−1	1.5822e−1	1.7475e−1	1.7007e−1	1.7288e−1	1.6920e−1
		(1.67e−2) −	(2.05e−1) −	(9.61e−4) +	(2.10e−3) +	(1.04e−3) +	(5.42e−3) −	(8.49e−4) ≈	(2.35e−3) −	(3.02e−3)
	5	6.2698e−1	1.9458e+0	5.0795e−1	4.9422e−1	4.7303e−1	5.7551e−1	5.1830e−1	4.7977e−1	5.5237e−1
		(1.25e−2) −	(2.91e−1) −	(1.87e−3) +	(7.10e−3) +	(9.91e−4) +	(2.18e−2) −	(1.55e−2) +	(7.72e−3) +	(2.26e−2)
	8	1.3644e+0	2.6011e+0	1.0845e+0	1.7865e+0	1.8763e+0	1.3816e+0	1.1378e+0	1.2964e+0	1.0539e+0
		(3.87e−2) −	(4.24e−1) −	(7.90e−3) −	(2.84e−1) −	(4.85e−1) −	(4.96e−2) −	(6.70e−2) −	(3.19e−2) −	(3.13e−2)
	10	1.9412e+0	5.0873e+0	1.3509e+0	2.2929e+0	2.9055e+0	1.9005e+0	1.4622e+0	1.5024e+0	1.3252e+0
		(8.07e−2) −	(3.83e+0) −	(4.09e−3) ≈	(1.26e−1) −	(9.03e−1) −	(6.54e−2) −	(1.21e−1) −	(1.23e−2) −	(7.77e−2)
	15	2.3155e+0	7.7070e+0	2.0512e+0	3.7182e+0	8.6313e+0	3.5141e+0	3.6591e+0	2.3468e+0	2.2457e+0
		(2.34e−1) ≈	(7.35e+0) −	(3.99e−2) +	(6.49e−1) −	(2.37e+0) −	(1.48e−1) −	(8.64e−1) −	(8.47e−3) ≈	(3.29e−1)
MaF12	3	3.0004e−1	2.1917e+0	2.2036e−1	2.2985e−1	2.2052e−1	2.3069e−1	2.2950e−1	2.2088e−1	2.0103e−1
		(8.98e−3) −	(5.69e−1) −	(2.77e−4) −	(2.98e−2) −	(2.32e−4) −	(3.05e−2) −	(2.89e−3) −	(4.04e−4) −	(1.33e−3)
	5	1.3610e+0	4.6976e+0	1.2141e+0	1.2161e+0	1.2139e+0	1.2071e+0	1.3625e+0	1.2192e+0	1.1060e+0
		(2.52e−2) −	(1.27e+0) −	(1.86e−3) −	(3.13e−3) −	(1.93e−3) −	(5.49e−3) −	(2.41e−2) −	(2.80e−3) −	(4.80e−3)
	8	3.4289e+0	1.1109e+1	3.5311e+0	3.5495e+0	3.5251e+0	3.5821e+0	3.5514e+0	3.5220e+0	3.2251e+0
		(4.36e−2) −	(2.66e+0) −	(1.04e−2) −	(1.45e−2) −	(1.72e−2) −	(3.10e−2) −	(2.34e−2) −	(2.51e−2) −	(2.48e−2)
	10	5.1842e+0	1.5212e+1	5.8112e+0	5.8747e+0	5.8312e+0	5.9181e+0	6.2083e+0	5.8335e+0	4.8281e+0
		(8.16e−2) −	(4.37e+0) −	(2.43e−2) −	(3.62e−2) −	(3.87e−2) −	(5.82e−2) −	(8.01e−2) −	(4.62e−2) −	(3.34e−2)
	15	9.1462e+0	2.5515e+1	1.1592e+1	1.2052e+1	1.1962e+1	1.2766e+1	1.1784e+1	1.1221e+1	9.4846e+0
		(9.54e−2) +	(7.06e+0) −	(3.17e−1) −	(5.67e−2) −	(1.88e−1) −	(2.39e−1) −	(5.67e−1) −	(4.32e−1) −	(1.08e−1)
MaF13	3	8.5605e−2	6.0225e−1	6.4233e−2	6.3039e−2	6.0801e−2	1.3238e−1	6.8878e−2	6.2834e−2	7.8803e−2
		(2.64e−3) −	(2.41e−1) −	(4.53e−3) +	(2.00e−3) +	(2.77e−3) +	(1.64e−2) −	(4.48e−3) +	(2.79e−3) +	(9.13e−3)
	5	1.0720e−1	7.9455e−1	1.4219e−1	1.8322e−1	4.4621e−1	3.8182e−1	3.1489e−1	4.3114e−1	1.0062e−1
		(8.39e−3) −	(7.92e−2) −	(5.83e−3) −	(1.02e−2) −	(5.95e−2) −	(1.09e−1) −	(2.84e−2) −	(2.69e−2) −	(7.51e−3)
	8	1.2289e−1	1.1440e+0	1.9096e−1	4.2561e−1	6.4498e−1	1.0589e+0	5.9450e−1	1.2387e+0	1.2225e−1
		(9.13e−3) ≈	(5.81e−2) −	(9.26e−3) −	(3.49e−2) −	(9.33e−2) −	(6.40e−1) −	(1.40e−1) −	(1.28e+0) −	(9.42e−3)
	10	1.2992e−1	1.2571e+0	2.2303e−1	5.5895e−1	7.3375e−1	1.9093e+0	1.0169e+0	1.0651e+0	1.2907e−1
		(8.66e−3) ≈	(1.54e−1) −	(1.94e−2) −	(5.25e−2) −	(9.08e−2) −	(3.59e+0) −	(1.90e−1) −	(2.48e−1) −	(6.95e−3)
	15	1.4160e−1	1.6269e+0	4.0537e−1	7.6555e−1	1.2150e+0	1.7355e+0	1.4065e+0	5.3653e+0	1.5057e−1
		(5.95e−3) +	(2.38e−1) −	(3.84e−2) −	(1.32e−1) −	(2.20e−1) −	(3.02e−1) −	(1.09e−1) −	(2.16e+1) −	(1.12e−2)
IDTLZ1	3	2.2388e−2	1.2345e−1	2.1744e−2	3.5132e−2	4.0429e−2	3.5932e−2	3.8869e−2	4.4658e−2	1.9962e−2
		(4.07e−4) −	(9.25e−2) −	(2.29e−5) −	(4.34e−5) −	(5.11e−4) −	(4.25e−4) −	(2.47e−4) −	(1.04e−2) −	(1.15e−4)
	5	7.4181e−2	1.7459e−1	7.5202e−2	1.1631e−1	3.7727e−1	1.0568e−1	1.2878e−1	1.5959e−1	6.7092e−2
		(9.47e−4) −	(1.18e−1) −	(1.78e−3) −	(3.02e−3) −	(7.77e−2) −	(2.79e−3) −	(1.06e−2) −	(2.13e−2) −	(4.98e−4)
	8	1.3140e−1	3.0139e−1	1.3253e−1	1.8263e−1	1.6467e−1	1.7573e−1	2.6989e−1	2.7704e−1	1.1839e−1
		(1.96e−3) −	(1.39e−1) −	(1.36e−3) −	(6.18e−2) −	(4.27e−3) −	(5.38e−3) −	(1.32e−2) −	(3.06e−2) −	(5.88e−3)
	10	1.5695e−1	3.5220e−1	1.5101e−1	2.2153e−1	1.9160e−1	1.8900e−1	2.8916e−1	3.2031e−1	1.2784e−1
		(2.76e−3) −	(1.57e−1) −	(7.36e−4) −	(1.17e−1) −	(1.05e−2) −	(1.33e−2) −	(8.22e−3) −	(4.26e−2) −	(2.40e−3)
	15	1.9501e−1	4.4624e−1	2.2088e−1	2.3610e−1	2.0667e−1	1.9751e−1	3.5219e−1	3.6402e−1	1.7252e−1
		(5.23e−3) ≈	(1.22e−1) −	(1.49e−2) −	(9.71e−2) −	(3.03e−2) −	(1.07e−2) −	(1.64e−2) −	(4.65e−2) −	(2.69e−2)

(*continued on next page*)

MaOEAIBP is similar to MaOEAIH, the distribution is not uniform, with larger gaps between individuals in the 0 to 2.5 range of objectives 6 to 10. Meanwhile, As an indicator-based MaOEA, MaOEAIGD struggles to maintain population diversity, thus obtaining the poorest result where all individuals converge at a single point. For the remaining algorithms, although their populations can cover the entire PF, these populations are not uniformly distributed and are mainly concentrated on the boundaries of the PF, leaving a large number of gaps in the

Table 4 (*continued*).

IDTLZ2	3	7.3682e−2	3.3274e−1	7.7457e−2	7.4857e−2	7.2105e−2	8.1145e−2	8.3395e−2	7.9669e−2	5.2057e−2
		(3.40e−3) −	(4.66e−3) −	(8.10e−5) −	(1.19e−3) −	(1.09e−3) −	(1.87e−3) −	(2.63e−3) −	(9.04e−4) −	(2.67e−4)
	5	2.3643e−1	4.9316e−1	2.3953e−1	2.5570e−1	3.3392e−1	2.7264e−1	3.3470e−1	3.6306e−1	1.9750e−1
		(4.17e−3) −	(1.53e−2) −	(5.07e−3) −	(3.51e−3) −	(2.67e−2) −	(2.03e−3) −	(1.99e−2) −	(1.09e−2) −	(1.03e−3)
	8	4.1933e−1	7.6873e−1	5.2711e−1	6.3207e−1	6.3923e−1	6.5265e−1	7.1084e−1	7.7445e−1	3.7413e−1
		(4.61e−3) −	(9.53e−3) −	(1.33e−2) −	(1.20e−2) −	(1.03e−2) −	(3.59e−3) −	(2.29e−2) −	(4.68e−2) −	(1.67e−3)
	10	4.7855e−1	7.9123e−1	5.2540e−1	7.2351e−1	7.5591e−1	7.3831e−1	7.5370e−1	7.4956e−1	4.5602e−1
		(3.50e−3) −	(2.43e−2) −	(1.22e−2) −	(1.25e−2) −	(1.25e−2) −	(3.19e−3) −	(2.15e−2) −	(1.82e−2) −	(2.00e−3)
	15	5.9912e−1	8.8669e−1	7.3698e−1	8.3192e−1	8.6624e−1	8.6286e−1	1.0123e+0	9.3189e−1	5.9153e−1
		(6.01e−3) −	(7.19e−4) −	(1.80e−2) −	(8.77e−3) −	(1.48e−2) −	(4.41e−3) −	(4.48e−2) −	(2.67e−2) −	(2.57e−3)
+/−/≈		7/59/9	0/73/2	10/62/3	4/69/2	6/68/1	6/66/3	6/67/1	8/64/3	

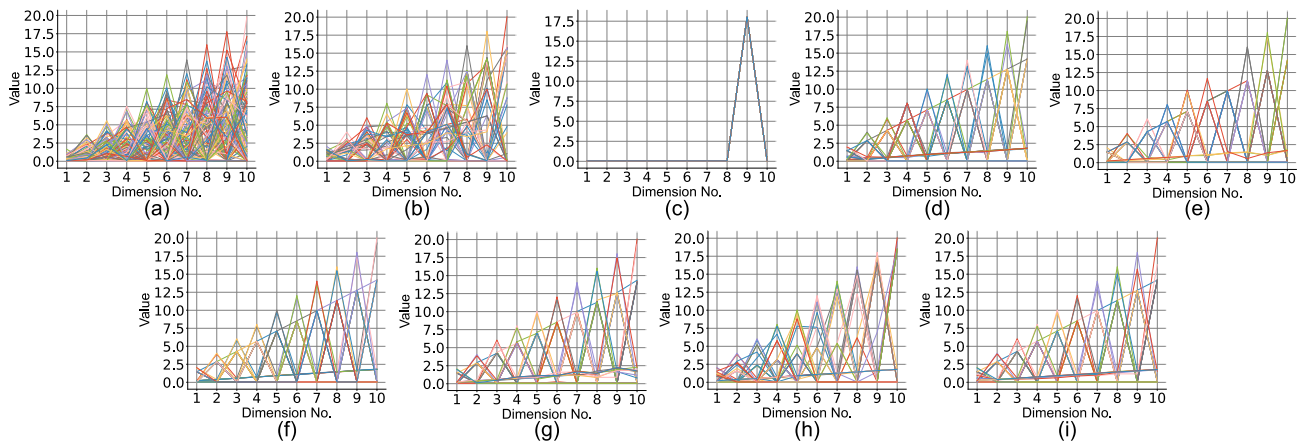

Fig. 11. Distribution of the final population for all algorithms on WFG5 problem with 10 objectives. The population obtained by MaOEAIH exhibits superior uniformity compared to other algorithms. (a)MaOEAIH. (b)MaOEAIBP. (c)MaOEAIGD. (d)ARMOEA. (e)HEA. (f)tDEA. (g)TSNSGAII. (h)EFRRR. (i)RVEA.

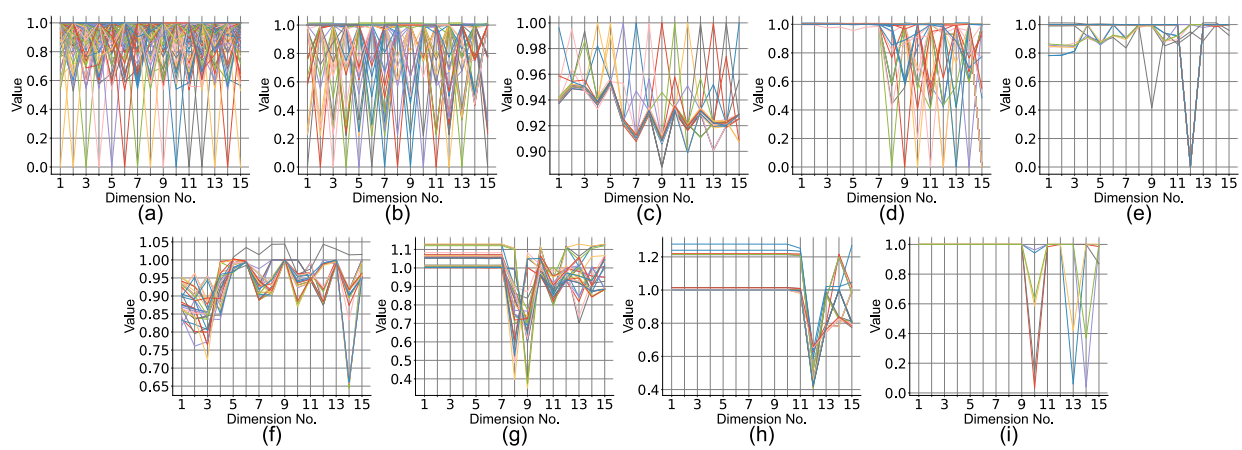

Fig. 12. Distribution of the final population for all algorithms on MAF1 problem with 15 objectives. The population obtained by MaOEAIH exhibits superior coverage and uniformity compared to other algorithms. (a)MaOEAIH. (b)MaOEAIBP. (c)MaOEAIGD. (d)ARMOEA. (e)HEA. (f)tDEA. (g)TSNSGAII. (h)EFRRR. (i)RVEA.

center of the PF. Overall, MaOEAIH is more effective at maintaining population diversity when dealing with MaOPs that have features like multimodality and large basin scales, compared to other algorithms.

Similarly, apart from achieving optimal results on 15-dimensional MAF4, where it is on par with MaOEAIBP, MaOEAIH also attained nearly all optimal results on MAF4 and MAF5 with severe scaling features. This is because GaRIF employs the principle of energy balance, enabling the population to uniformly spread across the entire PF regardless of how it is scaled. Consequently, MaOEAIH can effectively handle MaOPs with severe scaling features.

In the inversion problems of IDTLZ1, IDTLZ2, MAF1, and MAF2 under different objectives, MaOEAIH obtained 19 optimal results and 1 suboptimal result, demonstrating its effective handling of such problems. To visually demonstrate this result more intuitively, Fig. 12 shows the final population distribution of these algorithms under the 15-dimensional MAF1. From Fig. 12(e), it can be observed that although HEA enhanced selection pressure through hyper-dominance, its diversity maintenance method based on reference points led the population to converge to several points in the PF. Similarly, other algorithms using reference points also failed to maintain diversity. Although tDEA enhanced selection pressure through a new dominance relation, it lacked a diversity maintenance strategy, resulting in population diversity like algorithms using reference points. The population uniformity of MaOEAIGD is relatively good, but as shown in Fig. 12(c), the population distribution below the objective value of 0.94 is sparse. For MaOEAIBP, its diversity is better than the algorithms mentioned above, but as shown in Fig. 12(b), its distribution is sparse on the 14th objective. In contrast, as shown in Fig. 12(a), MaOEAIH maintains population diversity better when dealing with inversion problems, with a uniform and extensive population distribution.

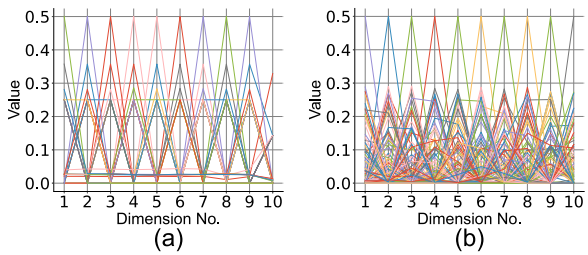

Fig. 13. Distribution of the final population for RVEA and RVEAGaRIF algorithms on the DTLZ1 problem with 10 objectives. The population obtained by (b)RVEAGaRIF is more uniform than that obtained by (a)RVEA.

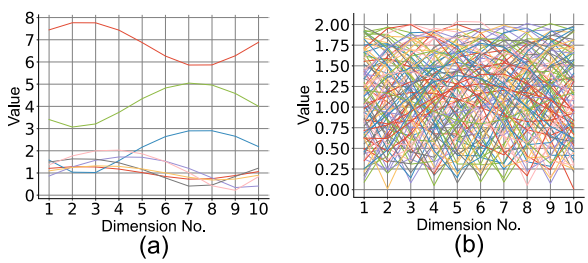

Fig. 14. Distribution of the final population for RVEA and RVEAGaRIF algorithms on the MAF8 problem with 10 objectives. The population obtained by (b)RVEAGaRIF exhibits superior convergence and diversity compared to the population obtained by (a)RVEA.

So far, we have demonstrated the effectiveness of MaOEA-IH for various MaOPs.

4.3. Effectiveness of GaRIF

To verify the effectiveness of weight vectors generated by MaOEAIH in handling MaOPs with regular PF and irregular PF, we replaced the weight vectors of RVEA with the weight vectors generated by MaOEAIH in the 31 test problems mentioned above. The algorithm with the replaced weight vectors is denoted as RVEAGaRIF. It is important to note that the weight vectors generated in MaOEAIH are extracted during the second use of the Pareto dominance-based evolution mechanism, and the ideal point for RVEAGaRIF was set as the coordinate origin. Subsequently, experiments were conducted on these test problems using RVEAGaRIF and RVEA.

Fig. 13 presents the parallel coordinate plot of the final populations obtained by RVEA and RVEAGaRIF on the 10-dimensional test problem DTLZ1, which has a regular PF. From the figure, it can be observed that both RVEAGaRIF and RVEA effectively converge to the PF. However, the final population obtained by RVEA does not cover the region of objective values between 0.05 and 0.25, whereas RVEAGaRIF can effectively cover these regions. This indicates that RVEAGaRIF exhibits better population diversity compared to RVEA. Fig. 14 depict the parallel coordinate plots of the final populations obtained by RVEA and RVEAGaRIF on the 10-dimensional test problem MAF8, which has irregular PF. In MAF8, the population obtained by RVEA fails to converge to the PF, while only retaining a small number of individuals in the population. However, the population obtained by RVEAGaRIF not only converges to the PF but also effectively covers the entire PF. This phenomenon is caused by the mismatch between the weight vectors used by RVEA and the shape of the PF, resulting in the waste of many weight vectors. As a result, individuals can only converge towards the remaining weight vectors, ultimately leading to their clustering together and the inability to completely cover the entire PF.

Furthermore, due to space limitations, we only provide the statistical results of the comparison between RVEAGaRIF and RVEA using the IGD indicator on the 3, 5, 8, and 10-objective experimental test suites in Table 5. From Table 5, it can be observed that in the DTLZ, WFG, MAF, and IDTLZ problem series with regular and irregular PFs, RVEAGaRIF outperforms RVEA in 103 out of 124 cases, approximately 83.1%. This demonstrates the competitive advantage of the algorithm that utilizes the weight vectors generated by MaOEAIH compared to the original algorithm when dealing with MaOPs with regular and irregular PFs.

Table 5
The statistical results of the comparison between RVEAGaRIF and RVEA using IGD (+ , −, ≈ respectively represent that RVEAGaRIF is worse, better, and no significant difference).

Comparisons		RVEA	vs.	RVEAGaRIF
		+	−	≈
Test problems	DTLZ1	1	3	0
	DTLZ2	0	4	0
	DTLZ3	0	4	0
	DTLZ4	0	4	0
	DTLZ5	0	3	1
	DTLZ6	0	4	0
	DTLZ7	0	4	0
	WFG1	2	2	0
	WFG2	1	3	0
	WFG3	0	4	0
	WFG4	1	3	0
	WFG5	1	3	0
	WFG6	2	2	0
	WFG7	1	3	0
	WFG8	1	3	0
	WFG9	1	3	0
	MAF1	0	4	0
	MAF2	0	4	0
	MAF3	0	4	0
	MAF4	0	4	0
	MAF5	2	2	0
	MAF6	0	4	0
	MAF7	0	4	0
	MAF8	0	4	0
	MAF9	2	2	0
	MAF10	2	2	0
	MAF11	1	3	0
	MAF12	1	3	0
	MAF13	1	3	0
	IDTLZ1	0	4	0
	IDTLZ2	0	4	0
$No.(M)$	3	3	28	0
	5	4	27	0
	8	4	27	0
	10	9	21	1
	All	20	103	1

4.4. Analysis of perturbation strategy

To investigate the impact of perturbation strategies on algorithm performance, we compared MaOEAIH with and without perturbation strategies. The experimental results are presented in Figs. 15 and 16. Fig. 15(a) displays a potential population obtained by the unperturbed MaOEAIH on the 3-dimensional DTLZ4 test problems. It can be observed that this population is concentrated on one boundary of the PF. In contrast, Fig. 15(b) shows the population obtained by the MaOEAIH with perturbation strategy, which covers the entire PF. This phenomenon may be attributed to the significant density differences in the point sets of the DTLZ4 test problem's PF. The population is likely to converge to areas with higher point density, unable to spread to regions with lower density, resulting in being trapped in local optima. This demonstrates that appropriate random perturbation helps the population spread into low-density regions, enabling escape from local optima. Fig. 16 illustrates the final populations obtained by the unperturbed and perturbed MaOEAIH strategies on the 10-dimensional DTLZ6 test problem. In Fig. 16(a), the unperturbed MaOEAIH fails to converge to the PF, while in Fig. 16(b), the perturbed MaOEAIH

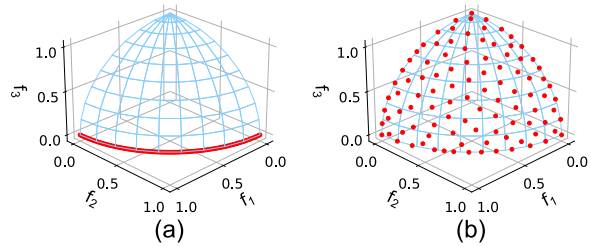

Fig. 15. Distribution of the final population for unperturbed and perturbed MaOEAIH on the DTLZ4 problem with 3 objectives. The population obtained by perturbed MaOEAIH demonstrates superior diversity compared to the population obtained by unperturbed MaOEAIH.

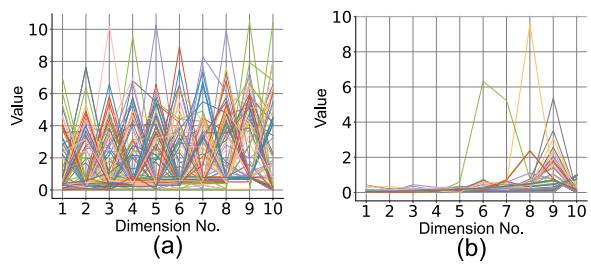

Fig. 16. Distribution of the final population for unperturbed and perturbed MaOEAIH on the DTLZ6 problem with 10 objectives. The population obtained by perturbed MaOEAIH demonstrates superior convergence compared to the population obtained by Unperturbed MaOEAIH.

achieves a population that approximately converges to the PF. If we appropriately increase the maximum number of evaluations, the unperturbed MaOEAIH strategy can also achieve a reasonable convergence to the PF. This indicates that perturbation strategies can improve the convergence rate of our algorithm on problems similar to DTLZ6.

4.5. The effect of eliminating dominant resistance solutions

To evaluate the effectiveness of the quartile method based on boundary solutions in eliminating dominance resistance solutions, we conducted comparative experiments between MaOEAIH and its variant without the quartile method based on boundary solutions, referred to as MaOEAIHDRS. We used a set of test suites with 15 objectives for the experiments. The experimental results, shown in Table 6, indicate that out of the 31 test problems, MaOEAIH significantly outperformed MaOEAIHDRS in 16 problems. This improvement can be attributed to the effective elimination of domination resistance solutions through the quartile method based on boundary solutions, particularly in difficult convergence problems like DTLZ1. This prevented a significant decrease in solution convergence when MaOEAIH pursued diversity. Additionally, 12 results were approximately comparable to MaOEAIHDRS. However, as the number of objectives increased, resulting in more domination resistance solutions, MaOEAIH might outperform MaOEAIHDRS in such cases. The three problems where MaOEAIH performed worse than MaOEAIHDRS were likely due to imprecise selection of boundary solutions using the coarse method, resulting in an incorrect estimation of the PF range and inadvertent elimination of useful solutions. This compromise compromised the algorithm's performance.

4.6. Research on optimization mechanism transformation parameters

The specific value for "a certain number of generations" in the context of "the minimum historical value of GaRIF of the population remains unchanged for a certain number of generations or when the minimum historical value of the sum of distances from each individual in the population to the origin remains unchanged" from Section 3.3 is determined for the following experiments. We conducted tests to evaluate the impact of different values for the "certain number of generations" on the algorithm and obtained the maximum "certain number of generations" required for three optimization mechanism transitions that yielded the overall best performance. Fig. 17(a) illustrates the source of the maximum "certain number of generations" required for the first transition, which was obtained from the 10-dimensional test problem WFG1. As the "certain number of generations" increases, the IGD value generally decreases, with a minimal decrease observed between 80 and 100 generations. Therefore, we set the "certain number of generations" for the first transition to 100. Fig. 17(b) presents the source of the maximum "certain number of generations" required for the second transition, which was obtained from the 10-dimensional test problem MAF6. Increasing the "certain number of generations" reduces the IGD value, but after 20 generations, the IGD value starts to increase. Hence, we set the "certain number of generations" for the second transition to 20. Fig. 17(c) displays the source of the maximum "certain number of generations" required for the third transition, obtained from the 10-dimensional test problem WFG3. The IGD value reaches its minimum at 60 generations. Therefore, we set the "certain number of generations" for the third transition to 60. It is important to note that in most test problems, such large values for the "certain number of generations" are not required, and only a few generations are sufficient. The value we selected is to balance all test problems, but due to the complexity and diversity of real-world problems, these values are rough estimates rather than precise ones. However, we chose them to maximize the possibility of producing superior results.

Table 6
Mean and standard deviation of the IGD values obtained by MaOEAIH and MaOEAIHDRS for DTLZ, WFG, MAF, and IDTLZ test suits.

Problem	*M*	MaOEAIHDRS	MaOEAIH
DTLZ1	15	1.8273e−1 (9.57e−3) −	1.5804e−1 (8.18e−3)
DTLZ2	15	5.9325e−1 (2.16e−3) +	5.9439e−1 (1.72e−3)
DTLZ3	15	6.1146e−1 (9.63e−3) −	6.0326e−1 (9.66e−3)
DTLZ4	15	6.0427e−1 (2.17e−3) −	5.9563e−1 (1.84e−3)
DTLZ5	15	1.1041e−1 (4.64e−1) −	4.8561e−2 (2.60e−2)
DTLZ6	15	3.8671e−2 (2.24e−2) ≈	3.9805e−2 (1.92e−2)
DTLZ7	15	1.6754e+0 (3.36e−2) −	1.5673e+0 (1.25e−2)
WFG1	15	1.7884e+0 (2.24e−1) ≈	1.8531e+0 (2.24e−1)
WFG2	15	2.2829e+0 (3.10e−1) ≈	2.1733e+0 (2.57e−1)
WFG3	15	4.3559e+0 (8.08e−1) ≈	4.2414e+0 (9.47e−1)
WFG4	15	9.4612e+0 (4.68e−2) ≈	9.4634e+0 (4.05e−2)
WFG5	15	9.4456e+0 (5.00e−2) −	9.4201e+0 (4.92e−2)
WFG6	15	9.4659e+0 (6.28e−2) ≈	9.4432e+0 (5.11e−2)
WFG7	15	9.4975e+0 (5.71e−2) ≈	9.4914e+0 (6.57e−2)
WFG8	15	9.6968e+0 (1.09e−1) −	9.6363e+0 (1.09e−1)
WFG9	15	9.5457e−1 (1.27e−1) −	9.5022e−1 (1.06e−1)
MaF1	15	3.2362e−1 (4.56e−3) −	3.0401e−1 (9.35e−4)
MaF2	15	2.3414e−1 (4.18e−3) ≈	2.3376e−1 (4.78e−3)
MaF3	15	2.0160e−1 (1.61e−2) −	1.5026e−1 (1.16e−2)
MaF4	15	2.0745e+3 (1.63e+2) +	2.4019e+3 (3.69e+2)
MaF5	15	2.1064e+3 (1.07e+2) −	1.9825e+3 (1.30e+2)
MaF6	15	6.9896e−2 (7.47e−2) ≈	9.4281e−2 (9.07e−2)
MaF7	15	1.6638e+0 (3.59e−2) −	1.5692e+0 (1.29e−2)
MaF8	15	2.0203e−1 (3.00e−3) −	1.9907e−1 (1.40e−3)
MaF9	15	2.5064e−1 (5.28e−2) −	2.2225e−1 (5.23e−2)
MaF10	15	1.8181e+0 (2.41e−1) ≈	1.8487e+0 (2.17e−1)
MaF11	15	2.3520e+0 (3.44e−1) ≈	2.2457e+0 (3.29e−1)
MaF12	15	9.5140e+0 (6.69e−2) ≈	9.4846e+0 (1.08e−1)
MaF13	15	1.6494e−1 (1.11e−2) −	1.5057e−1 (1.12e−2)
IDTLZ1	15	1.8292e−1 (2.23e−2) −	1.7252e−1 (2.69e−2)
IDTLZ2	15	5.7985e−1 (3.46e−3) +	5.9153e−1 (2.57e−3)
+/ − / ≈		3/16/12	

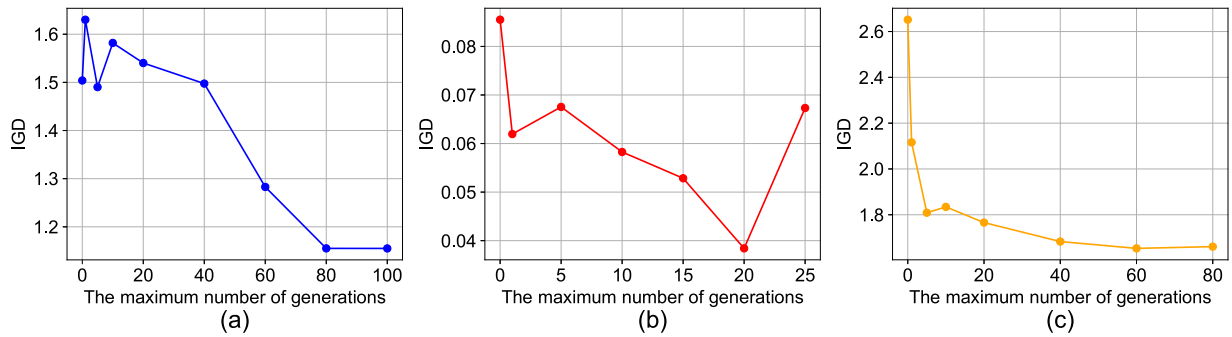

Fig. 17. The source of the maximum "certain number of generations". A smaller IGD value indicates that "certain number of generations" is more appropriate. (a)First transition in 10-dimensional WFG1. (b)Second transition in 10-dimensional MAF6. (c)Last transition in 10-dimensional WFG3.

Table 7
Running time of all algorithms.

Algorithm	MaOEAIBP	MaOEAIGD	ARMOEA	HEA	tDEA	TSNSGAII	EFRRR	RVEA	MaOEAIH
Total time	11 265.44	8456.31	46 211.385	6727.249	5194.235	11 469.973	6214.766	3484.68	52 763.07
Rank	6	5	8	4	2	7	3	1	9

4.7. Comparison of algorithms' running times

To evaluate the efficiency of MaOEAIH in handling MaOPs, this section summarizes the total time required by MaOEAIH and its comparative algorithms to process all test problems (measured in seconds), as shown in Table 7. The results indicate that RVEA has the shortest runtime. In contrast, MaOEAIBP, ARMOEA, TSNSGAII, and MaOEAIH take much longer, all exceeding 10^4 seconds, with MaOEAIH taking the longest because selecting the population with the minimum GaRIF requires a significant amount of time. Despite MaOEAIH's higher time complexity, its time consumption is acceptable given its superior performance.

In summary, MaOEAIH is suitable for solving problems that are complex but not extremely time-sensitive. From another perspective, it is also applicable to scenarios where the objective function evaluation is very time-consuming because, in such cases, the evaluation time of the objective function far exceeds the algorithm running time, making the running time differences between algorithms negligible. For example, in real-world applications like vehicle structure design in automotive crash safety [64], and pipeline systems with a large number of decision variables [65], rapid response may not be the primary concern, but improvements in the quality of the solution set can have a more significant impact.

5. Conclusion

In this paper, a many-objective evolutionary algorithm based on interaction force and hybrid optimization mechanism (MaOEAIH) is proposed to balance convergence and diversity. In MaOEAIH, a population selected based on Pareto dominance and the Gravity and Repulsion-Interaction Force is employed as the weight vectors in the decomposition-based optimization mechanism to obtain a population with good convergence and diversity. Additionally, a quartile method based on boundary solutions is used to maintain convergence, and a random perturbation method is applied to help the population escape local optima.

Comparing with 8 state-of-the-art algorithms on 31 test problems, MaOEAIH shows competitive performance compared to other algorithms. Furthermore, experiments conducted on the weight vectors generated by MaOEAIH show that these weight vectors effectively handle problems with both regular and irregular Pareto fronts. The experiments on the methods for eliminating dominance resistance solutions and the perturbation strategy also validate their effectiveness.

When dealing with most MaOPs, MaOEAIH can effectively improve the diversity and convergence of the population, enhance the selection pressure on the population, maintain a balance between convergence and diversity, and help the population escape local optima. However, MaOEAIH lacks the means to handle MaOPs with convex PF, leading to relatively poor results in these problems. Additionally, MaOEAIH spends a considerable amount of time when using GaRIF to select individuals, resulting in a higher time complexity. Finally, MaOEAIH only undergoes performance verification on the test suites, and its effectiveness in real-world applications remains to be studied. In future work, we will focus on researching a method that can accurately identify boundary solutions to handle problems with convex PF, while also using determinant point processes to reduce the selection time of GaRIF. Furthermore, recent constrained MaOPs have received widespread attention, and MaOEAIH will be applied to constrained real-world applications such as unmanned aircraft path detection to enhance its practical value. However, MaOEAIH was not designed with constraint problems in mind, making it challenging to improve MaOEAIH to handle constraint problems. At the same time, tuning certain parameters of MaOEAIH in handling real-world applications is also a challenge.

CRediT authorship contribution statement

Lei Yang: Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing. **Jiale Cao:** Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Kangshun Li:** Funding acquisition, Supervision. **Yuanye Zhang:** Investigation, Visualization, Writing – review & editing. **Rui Xu:** Investigation, Visualization. **Ke Li:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.swevo.2024.101667.

References

[1] J. Liu, L. Xu, Y. Hu, H. Jahanshahi, Flow measurement data quality improvement-oriented optimal flow sensor configuration, Swarm Evol. Comput. 80 (2023) 101325, http://dx.doi.org/10.1016/j.swevo.2023.101325.

[2] A. Salih, A. Moshaiov, Promoting transfer of robot neuro-motion-controllers by many-objective topology and weight evolution, IEEE Trans. Evol. Comput. 27 (2) (2023) 385–395, http://dx.doi.org/10.1109/TEVC.2022.3172294.

[3] S.-C. Liu, Z.-G. Chen, Z.-H. Zhan, S.-W. Jeon, S. Kwong, J. Zhang, Many-objective job-shop scheduling: A multiple populations for multiple objectives-based genetic algorithm approach, IEEE Trans. Cybern. 53 (3) (2023) 1460–1474, http://dx.doi.org/10.1109/TCYB.2021.3102642.

[4] L. Miguel Antonio, C.A. Coello Coello, Coevolutionary multiobjective evolutionary algorithms: Survey of the state-of-the-art, IEEE Trans. Evol. Comput. 22 (6) (2018) 851–865, http://dx.doi.org/10.1109/TEVC.2017.2767023.

[5] C.A. Coello Coello, S. González Brambila, J. Figueroa Gamboa, M.G. Castillo Tapia, R. Hernández Gómez, Evolutionary multiobjective optimization: open research areas and some challenges lying ahead, Complex Intell. Syst. 6 (2) (2020) 221–236, http://dx.doi.org/10.1007/s40747-019-0113-4.

[6] M. Farina, P. Amato, On the optimal solution definition for many-criteria optimization problems, in: 2002 Annual Meeting of the North American Fuzzy Information Processing Society Proceedings, IEEE, 2002, pp. 233–238, http://dx.doi.org/10.1109/NAFIPS.2002.1018061.

[7] B. Li, J. Li, K. Tang, X. Yao, Many-objective evolutionary algorithms: A survey, ACM Comput. Surv. 48 (1) (2015) 1–35, http://dx.doi.org/10.1145/2792984.

[8] X. Wang, Z. Xie, X. Zhou, X. Gu, A two-stage adaptive reference direction guided evolutionary algorithm with modified dominance relation for many-objective optimization, Swarm Evol. Comput. 78 (2023) 101272, http://dx.doi.org/10.1016/j.swevo.2023.101272.

[9] T. Liu, Z. Wang, M. Wei, A many-objective optimization algorithm using a two-space interactive evolutionary framework, Swarm Evol. Comput. 75 (2022) 101185, http://dx.doi.org/10.1016/j.swevo.2022.101185.

[10] Z. Mao, M. Liu, A local search-based many-objective five-element cycle optimization algorithm, Swarm Evol. Comput. 68 (2022) 101009, http://dx.doi.org/10.1016/j.swevo.2021.101009.

[11] S. Zhu, L. Xu, E.D. Goodman, Z. Lu, A new many-objective evolutionary algorithm based on generalized Pareto dominance, IEEE Trans. Cybern. 52 (8) (2022) 7776–7790, http://dx.doi.org/10.1109/TCYB.2021.3051078.

[12] J. Shen, P. Wang, X. Wang, A controlled strengthened dominance relation for evolutionary many-objective optimization, IEEE Trans. Cybern. 52 (5) (2022) 3645–3657, http://dx.doi.org/10.1109/TCYB.2020.3015998.

[13] G. Yu, Y. Jin, M. Olhofer, A multiobjective evolutionary algorithm for Finding Knee Regions using two localized dominance relationships, IEEE Trans. Evol. Comput. 25 (1) (2021) 145–158, http://dx.doi.org/10.1109/TEVC.2020.3008877.

[14] Y. Xu, H. Zhang, X. Zeng, Y. Nojima, An adaptive convergence enhanced evolutionary algorithm for many-objective optimization problems, Swarm Evol. Comput. 75 (2022) 101180, http://dx.doi.org/10.1016/j.swevo.2022.101180.

[15] Y. Zhou, S. Li, W. Pedrycz, G. Feng, ACDB-EA: Adaptive convergence-diversity balanced evolutionary algorithm for many-objective optimization, Swarm Evol. Comput. 75 (2022) 101145, http://dx.doi.org/10.1016/j.swevo.2022.101145.

[16] G. Chen, J. Li, A diversity ranking based evolutionary algorithm for multi-objective and many-objective optimization, Swarm Evol. Comput. 48 (2019) 274–287, http://dx.doi.org/10.1016/j.swevo.2019.03.009.

[17] Q. Zhang, H. Li, MOEA/D: A multiobjective evolutionary algorithm based on decomposition, IEEE Trans. Evol. Comput. 11 (6) (2007) 712–731, http://dx.doi.org/10.1109/TEVC.2007.892759.

[18] D.K. Saxena, K. Deb, Non-linear dimensionality reduction procedures for certain large-dimensional multi-objective optimization problems: Employing correntropy and a novel maximum variance unfolding, in: Evolutionary Multi-Criterion Optimization, Springer, 2007, pp. 772–787, http://dx.doi.org/10.1007/978-3-540-70928-2_58.

[19] H. Ishibuchi, Y. Setoguchi, H. Masuda, Y. Nojima, Performance of decomposition-based many-objective algorithms strongly depends on Pareto front shapes, IEEE Trans. Evol. Comput. 21 (2) (2017) 169–190, http://dx.doi.org/10.1109/TEVC.2016.2587749.

[20] H. Ishibuchi, R. Imada, N. Masuyama, Y. Nojima, Use of two reference points in hypervolume-based evolutionary multiobjective optimization algorithms, in: International Conference on Parallel Problem Solving from Nature, Springer, 2018, pp. 384–396, http://dx.doi.org/10.1007/978-3-319-99253-2_31.

[21] M. Li, X. Yao, What weights work for you? Adapting weights for any Pareto front shape in decomposition-based evolutionary multi-objective optimisation, Evol. Comput. 28 (2) (2017) 227–253, http://dx.doi.org/10.1162/evco_a_00269.

[22] Y. Qi, X. Ma, F. Liu, L. Jiao, J. Sun, J. Wu, MOEA/D with adaptive weight adjustment, Evol. Comput. 22 (2) (2014) 231–264, http://dx.doi.org/10.1162/EVCO_a_00109.

[23] S. Liu, Q. Lin, K.-C. Wong, C.A. Coello Coello, J. Li, Z. Ming, J. Zhang, A self-guided reference vector strategy for many-objective optimization, IEEE Trans. Cybern. 52 (2) (2022) 1164–1178, http://dx.doi.org/10.1109/TCYB.2020.2971638.

[24] C. Zhao, Y. Zhou, Y. Hao, Decomposition-based evolutionary algorithm with dual adjustments for many-objective optimization problems, Swarm Evol. Comput. 75 (2022) 101168, http://dx.doi.org/10.1016/j.swevo.2022.101168.

[25] L.R. de Farias, A.F. Araújo, A decomposition-based many-objective evolutionary algorithm updating weights when required, Swarm Evol. Comput. 68 (2022) 100980, http://dx.doi.org/10.1016/j.swevo.2021.100980.

[26] J. Bader, E. Zitzler, Hype: An algorithm for fast hypervolume-based many-objective optimization, Evol. Comput. 19 (1) (2011) 45–76, http://dx.doi.org/10.1162/EVCO_a_00009.

[27] Y. Sun, G.G. Yen, Z. Yi, IGD indicator-based evolutionary algorithm for many-objective optimization problems, IEEE Trans. Evol. Comput. 23 (2) (2019) 173–187, http://dx.doi.org/10.1109/TEVC.2018.2791283.

[28] K. Shang, H. Ishibuchi, A new hypervolume-based evolutionary algorithm for many-objective optimization, IEEE Trans. Evol. Comput. 24 (5) (2020) 839–852, http://dx.doi.org/10.1109/TEVC.2020.2964705.

[29] L. While, P. Hingston, L. Barone, S. Huband, A faster algorithm for calculating hypervolume, IEEE Trans. Evol. Comput. 10 (1) (2006) 29–38, http://dx.doi.org/10.1109/TEVC.2005.851275.

[30] Q. Zhang, A. Zhou, Y. Jin, RM-MEDA: A regularity model-based multiobjective estimation of distribution algorithm, IEEE Trans. Evol. Comput. 12 (1) (2008) 41–63, http://dx.doi.org/10.1109/TEVC.2007.894202.

[31] D. Brockhoff, T. Wagner, H. Trautmann, On the properties of the R2 indicator, in: Proceedings of the 14th Annual Conference on Genetic and Evolutionary Computation, ACM, 2012, pp. 465–472, http://dx.doi.org/10.1145/2330163.2330230.

[32] J. Zhang, L. Wei, R. Fan, H. Sun, Z. Hu, Solve large-scale many-objective optimization problems based on dual analysis of objective space and decision space, Swarm Evol. Comput. 70 (2022) 101045, http://dx.doi.org/10.1016/j.swevo.2022.101045.

[33] H. Chen, R. Cheng, J. Wen, H. Li, J. Weng, Solving large-scale many-objective optimization problems by covariance matrix adaptation evolution strategy with scalable small subpopulations, Inform. Sci. 509 (2020) 457–469, http://dx.doi.org/10.1016/j.ins.2018.10.007.

[34] Q. Liu, J. Zou, S. Yang, J. Zheng, A multiobjective evolutionary algorithm based on decision variable classification for many-objective optimization, Swarm Evol. Comput. 73 (2022) 101108, http://dx.doi.org/10.1016/j.swevo.2022.101108.

[35] W. Du, M. Zhang, W. Ying, M. Perc, K. Tang, X. Cao, D. Wu, The networked evolutionary algorithm: A network science perspective, Appl. Math. Comput. 338 (2018) 33–43, http://dx.doi.org/10.1016/j.amc.2018.06.002.

[36] I. Fister, A. Iglesias, A. Galvez, J. Del Ser, E. Osaba, I. Fister, M. Perc, M. Slavinec, Novelty search for global optimization, Appl. Math. Comput. 347 (2019) 865–881, http://dx.doi.org/10.1016/j.amc.2018.11.052.

[37] Y. Liu, N. Zhu, M. Li, Solving many-objective optimization problems by a Pareto-based evolutionary algorithm with preprocessing and a penalty mechanism, IEEE Trans. Cybern. 51 (11) (2021) 5585–5594, http://dx.doi.org/10.1109/TCYB.2020.2988896.

[38] K. Li, K. Deb, Q. Zhang, S. Kwong, An evolutionary many-objective optimization algorithm based on dominance and decomposition, IEEE Trans. Evol. Comput. 19 (5) (2015) 694–716, http://dx.doi.org/10.1109/TEVC.2014.2373386.

[39] A. Ikai, Chapter four - interaction forces, in: The World of Nano-Biomechanics (Second Edition), Elsevier, 2017, pp. 63–76, http://dx.doi.org/10.1016/B978-0-444-63686-7.00004-3.

[40] D. Hardin, E. Saff, Minimal Riesz energy point configurations for rectifiable d-dimensional manifolds, Adv. Math. 193 (1) (2005) 174–204, http://dx.doi.org/10.1016/j.aim.2004.05.006.

[41] J.G. Falcón-Cardona, H. Ishibuchi, C.A.C. Coello, Riesz s-energy-based reference sets for multi-objective optimization, in: 2020 IEEE Congress on Evolutionary Computation, IEEE, 2020, pp. 1–8, http://dx.doi.org/10.1109/CEC48606.2020.9185833.

[42] J. Blank, K. Deb, Y. Dhebar, S. Bandaru, H. Seada, Generating well-spaced points on a unit simplex for evolutionary many-objective optimization, IEEE Trans. Evol. Comput. 25 (1) (2021) 48–60, http://dx.doi.org/10.1109/TEVC.2020.2992387.

[43] Y. Liu, D. Gong, J. Sun, Y. Jin, A many-objective evolutionary algorithm using a one-by-one selection strategy, IEEE Trans. Cybern. 47 (9) (2017) 2689–2702, http://dx.doi.org/10.1109/TCYB.2016.2638902.

[44] C.C. Aggarwal, A. Hinneburg, D.A. Keim, On the surprising behavior of distance metrics in high dimensional space, in: International Conference on Database Theory, Springer, 2001, pp. 420–434, http://dx.doi.org/10.1007/3-540-44503-X_27.

[45] K. Deb, Multi-objective optimisation using evolutionary algorithms: An introduction, in: Multi-Objective Evolutionary Optimisation for Product Design and Manufacturing, Springer London, 2011, pp. 3–34, http://dx.doi.org/10.1007/978-0-85729-652-8_1.

[46] H.-G. Beyer, H.-P. Schwefel, Evolution strategies –a comprehensive introduction, Nat. Comput. 1 (1) (2002) 3–52, http://dx.doi.org/10.1023/A:1015059928466.

[47] K. Deb, Geneas: A robust optimal design technique for mechanical component design, in: Evolutionary Algorithms in Engineering Applications, Springer Berlin Heidelberg, 1997, pp. 497–514, http://dx.doi.org/10.1007/978-3-662-03423-1_27.

[48] Z. Liang, T. Luo, K. Hu, X. Ma, Z. Zhu, An indicator-based many-objective evolutionary algorithm with boundary protection, IEEE Trans. Cybern. 51 (9) (2021) 4553–4566, http://dx.doi.org/10.1109/TCYB.2019.2960302.

[49] Y. Tian, R. Cheng, X. Zhang, F. Cheng, Y. Jin, An indicator-based multiobjective evolutionary algorithm with reference point adaptation for better versatility, IEEE Trans. Evol. Comput. 22 (4) (2018) 609–622, http://dx.doi.org/10.1109/TEVC.2017.2749619.

[50] Z. Liu, F. Han, Q. Ling, H. Han, J. Jiang, A many-objective optimization evolutionary algorithm based on hyper-dominance degree, Swarm Evol. Comput. 83 (2023) 101411, http://dx.doi.org/10.1016/j.swevo.2023.101411.

[51] Y. Yuan, H. Xu, B. Wang, X. Yao, A new dominance relation-based evolutionary algorithm for many-objective optimization, IEEE Trans. Evol. Comput. 20 (1) (2016) 16–37, http://dx.doi.org/10.1109/TEVC.2015.2420112.

[52] F. Ming, W. Gong, L. Wang, A two-stage evolutionary algorithm with balanced convergence and diversity for many-objective optimization, IEEE Trans. Syst. Man Cybern.: Syst. 52 (10) (2022) 6222–6234, http://dx.doi.org/10.1109/TSMC.2022.3143657.

[53] Y. Yuan, H. Xu, B. Wang, B. Zhang, X. Yao, Balancing convergence and diversity in decomposition-based many-objective optimizers, IEEE Trans. Evol. Comput. 20 (2) (2016) 180–198, http://dx.doi.org/10.1109/TEVC.2015.2443001.

[54] R. Cheng, Y. Jin, M. Olhofer, B. Sendhoff, A reference vector guided evolutionary algorithm for many-objective optimization, IEEE Trans. Evol. Comput. 20 (5) (2016) 773–791, http://dx.doi.org/10.1109/TEVC.2016.2519378.

[55] Y. Tian, R. Cheng, X. Zhang, Y. Jin, Platemo: A MATLAB platform for evolutionary multi-objective optimization [educational forum], IEEE Comput. Intell. Mag. 12 (4) (2017) 73–87, http://dx.doi.org/10.1109/MCI.2017.2742868.

[56] K. Deb, L. Thiele, M. Laumanns, E. Zitzler, Scalable test problems for evolutionary multiobjective optimization, in: Evolutionary Multiobjective Optimization: Theoretical Advances and Applications, Springer London, 2005, pp. 105–145, http://dx.doi.org/10.1007/1-84628-137-7_6.

[57] S. Huband, P. Hingston, L. Barone, L. While, A review of multiobjective test problems and a scalable test problem toolkit, IEEE Trans. Evol. Comput. 10 (5) (2006) 477–506, http://dx.doi.org/10.1109/TEVC.2005.861417.

[58] R. Cheng, M. Li, Y. Tian, X. Zhang, S. Yang, Y. Jin, X. Yao, A benchmark test suite for evolutionary many-objective optimization, Complex Intell. Syst. 3 (1) (2017) 67–81, http://dx.doi.org/10.1007/s40747-017-0039-7.

[59] H. Jain, K. Deb, An evolutionary many-objective optimization algorithm using reference-point based nondominated sorting approach, part II: Handling constraints and extending to an adaptive approach, IEEE Trans. Evol. Comput. 18 (4) (2014) 602–622, http://dx.doi.org/10.1109/TEVC.2013.2281534.

[60] O. Schutze, X. Esquivel, A. Lara, C.A. Coello Coello, Using the averaged hausdorff distance as a performance measure in evolutionary multiobjective optimization, IEEE Trans. Evol. Comput. 16 (4) (2012) 504–522, http://dx.doi.org/10.1109/TEVC.2011.2161872.

[61] Y. Cui, Z. Geng, Q. Zhu, Y. Han, Review: Multi-objective optimization methods and application in energy saving, Energy 125 (2017) 681–704, http://dx.doi.org/10.1016/j.energy.2017.02.174.

[62] H. Wang, Y. Jin, X. Yao, Diversity assessment in many-objective optimization, IEEE Trans. Cybern. 47 (6) (2017) 1510–1522, http://dx.doi.org/10.1109/TCYB.2016.2550502.

[63] J. Derrac, S. García, D. Molina, F. Herrera, A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms, Swarm Evol. Comput. 1 (1) (2011) 3–18, http://dx.doi.org/10.1016/j.swevo.2011.02.002.

[64] X. Liao, Q. Li, X. Yang, W. Zhang, W. Li, Multiobjective optimization for crash safety design of vehicles using stepwise regression model, Struct. Multidiscip. Optim. 35 (6) (2008) 561–569, http://dx.doi.org/10.1007/s00158-007-0163-x.

[65] S. Mahdavi, M.E. Shiri, S. Rahnamayan, Metaheuristics in large-scale global continues optimization: A survey, Inform. Sci. 295 (2015) 407–428, http://dx.doi.org/10.1016/j.ins.2014.10.042.